



BIONETS

WP 1.3 – PERFORMANCE ANALYSIS AND EVALUATION

D1.3.2 BIONETS simulation framework and initial performance analysis

Reference:	BIONETS/BUTE/wp1.3/1.3
Category:	Deliverable
Editor:	Sándor Szabó (BUTE)
Authors:	Vilmos Simon, Sándor Szabó, László Bacsárdi, Márton Bérces, Tibor Csvórics, Balázs Tujner, Endre Varga (BUTE), Francesco De Pellegrini, Daniele Miorandi, Iacopo Carreras (CN), A. Panagakis, A. Vaios (NKUA), Eleonora Borgia, Luciana Pelusi (CN-CNR), Nicolas Bonneau, Mérouane Debbah, Eitan Altman, Are Hjørungnes (INRIA), Juhani Latvakoski, Tomi Hautakoski (VTT)
Verification:	David Linner (TUB), Giovanni Neglia (INRIA)
Date:	Aug 5, 2007
Status:	Final
Availability:	Public

SUMMARY

Innovative mathematical models developed for the BIONETS environment are analyzed in this deliverable presenting a comprehensive performance evaluation for the first time in the BIONETS project, which will give an important feedback on the design of the system properties. The main focus is on the efficiency of the information dissemination algorithms with respect to reliability of these algorithms, scalability and induced overhead. The algorithms are compared by employing the BIONETS simulation platform, a simulation framework that will be used by every partner. The key objective of the simulator is to provide a tool for all partners, so they can test the implementations of early ideas and algorithms from any Work Package. The deliverable furthermore contains the theoretical studies carried out in WP1.3.1, as the work on the mathematical models continues in parallel with the performance evaluation of the earlier models.

Contents

1	Introduction	7
2	Connectivity and Scalability Issues	9
2.1	A connectivity graph	11
2.2	Diffusion and Bounds	12
2.2.1	Critical window and delay bound	13
2.3	Numerical Results	14
3	Network Design and Optimization: Clustered Infrastructure	18
3.1	Model and Notations	19
3.2	Matched Filter	19
3.2.1	Infinite Cluster Network	19
3.2.2	Joint Cluster Processing	20
3.3	Optimum receiver	20
3.3.1	Infinite Cluster Network	20
3.3.2	Joint Cluster Processing	20
3.4	Conclusion	20
4	Network Design and Optimization: Ad-Hoc Network Infrastructure	21
4.1	Introduction	21
4.2	System Model	22
4.2.1	T-nodes Clustering	22

4.2.2	Power Attenuation	22
4.3	Analysis	23
4.3.1	Signals Expressions in Time Domain	23
4.3.2	Asymptotic analysis for Capacity expression	23
4.4	Simulations and Results	24
4.5	Conclusion	24
5	Scalability under energy and bandwidth constraints	26
5.1	Introduction	26
5.2	Model	27
5.3	Games and Equilibria	27
5.4	Power Allocation in the Nash Equilibrium	28
5.5	Successive Interference Cancellation	28
5.6	Numerical Results	29
5.7	Conclusion	29
6	The BIONETS simulator platform	31
6.1	Benefits	31
6.2	Goals	31
6.3	Realization	31
6.4	The simulation framework	32
6.5	Main structure	32
6.6	Module Library architecture	33
6.7	Reference algorithms for information dissemination	34
6.8	Scalable Broadcast Algorithm (SBA)	35
6.9	Generic Self-Pruning	36
6.10	Hypergossiping	36
6.11	New algorithms developed in WP1.3	37
6.12	Multi-Message SBA (MMSBA) — An improved version of SBA	37
6.13	A new algorithm: Modified IOBIO (InfOrmation Dissemination Protocol for BIO- logically Inspired autonomic Networks and Services)	38
6.14	Simulation setup	39
6.15	Mobility models	40
6.15.1	Constant Speed Mobility	40
6.15.2	Reference Point Group Mobility Model (RPGMM)	40
6.15.3	Reference Point Group Mobility Model with Dynamic Groups (RPGMMDG)	41
6.16	Simulation results	41

7	Simulations of situation-aware opportunistic routing	47
7.1	Introduction and motivation	47
7.2	Model structure	48
7.3	Simulation scenarios	50
7.4	Simulations carried out and preliminary results obtained	50
7.5	Conclusions and future work	53
8	Message spreading in non-cooperative environments	55
8.1	Adjustability to non-cooperative environments	56
9	Tnode to Unode communications: from real-world observation to a realistic model for the BIONETS simulation framework	61
9.1	Motivation and Problem Statement	61
9.1.1	Application Scenarios	62
9.1.2	A novel adaptive communication paradigm based on Reed-Solomon codes	64
9.2	Tnode to Unode Communications Modelling	64
9.2.1	Tnode to Unode communications in the BIONETS simulation framework .	64
9.2.2	Trace-driven simulations	65

DOCUMENT HISTORY

Version History

Version	Status	Date	Author(s)
0.1	Draft	10 May 2007	Vilmos Simon (BUTE)
0.2	Draft	10 Jun 2007	László Bacsárdi (BUTE)
0.3	Draft	13 Jun 2007	László Bacsárdi (BUTE)
0.4	Draft	16 Jun 2007	László Bacsárdi (BUTE)
0.5	Draft	20 Jun 2007	Vilmos Simon (BUTE)
1.0	Final	21 Jun 2007	Vilmos Simon (BUTE)
1.1	Final	4 Jul 2007	Vilmos Simon (BUTE)
1.2	Draft	16 Jul 2007	László Bacsárdi (BUTE)
1.3	Draft	5 Aug 2007	László Bacsárdi (BUTE)

Summary of Changes

Version	Section(s)	Synopsis of Change
0.1	Outline	
0.2	Outline	BUTE part included
0.3	Outline	CN part included
0.4	Whole document	All partners' contribution included
0.5	Whole document	Final integration
1.0	Whole document	Revision, final document
1.1	Section 3, 4, 5	Chapters added
1.2	Whole document	Review 1
1.3	Whole document	Review 2

1 Introduction

In D1.3.1 [1] the first mathematical models developed for the BIONETS environment were introduced, focusing on mobility modeling and on specific mobility management schemes, examining the data diffusion in the system (IOBIO protocol, two-hop-relaying algorithms). Another issue was the M^2 mobility framework and the application of CDMA in BIONETS, by optimizing the radio transmission.

The innovative networking functionalities and algorithms presented in D1.3.1 [1] are analyzed in this deliverable presenting an exhaustive performance evaluation for the first time in the BIONETS project, which will provide a significant feedback towards the design of the system properties. The main focus is on the efficiency of the information dissemination algorithms with respect to reliability of these algorithms, scalability and induced overhead; therefore these different algorithms are compared by employing the BIONETS simulation platform. The vision of the simulator platform is to provide a simulation framework that will be used by every partner of the BIONETS. It will include the algorithms as well as the ideas from the entire BIONETS community. The main goal of the simulator is to provide a tool for all participants, so they can access realizations of previous concepts and algorithms from any Work Package. Another goal is to facilitate experimentation with novel ideas that can be incorporated into BIONETS. The simulator is designed to support quick prototyping and the swift realization of ideas. There is also a great emphasis on flexibility. The first part of the deliverable contains the theoretical studies carried out in Work Package 1.3.1, as the work on the mathematical models continues in parallel with the performance evaluation of the earlier models. In the theoretical section we introduce some definitions aimed at capturing the properties over time of connectivity in the U-nodes plane (Chapter 2). The main definition introduced here is the connectivity graph, which collects, to some extent, the information on the contact process of the U-nodes. The cumulative information on the contact process retained by the connectivity graph let us obtain scaling laws for the delay in delay tolerant networks (DTN), under uniformity assumptions on the mobility pattern. We validated our model comparing the typical parameters of the connectivity graph with the corresponding values of an Erdos-Renyi random graph using synthetic mobility and real-world traces.

Another important theoretical advance is a study on the capacity scaling of a system where capacity expressions accounting for physical characteristics of the environment (topology, frequency band, etc.) are provided and an asymptotic analysis is performed for the increasing size of the dense scattering environment (Chapter 3, 4). We studied how capacity scales when the size the network increases, as well as the point at which asymptotic regime is reached depending on the positioning of nodes. The theoretical part is concluded with the performance analysis of a multi-user system in the context of frequency selective fading channels (Chapter 5). Using game theoretic tools, a useful framework is provided in order to determine the optimal power allocation when T-nodes know only their own channel (while perfect channel state information is assumed at the U-node).

The second part of the deliverable is all about the simulation platform and the performance evaluation results obtained by the simulator. The details of the simulation framework realization,

the main structure and the Module Library architecture are introduced. We have used the simulator to evaluate the performance of different information dissemination algorithms and mobility models, presenting the two novel information dissemination algorithms, the Multi-Message SBA (MMSBA) and the Modified IOBIO (MIOBIO), comparing them with the reference algorithms chosen from the literature, which are the best available dissemination methods published earlier for a BIONETS environment (SBA, Generic Self-Pruning, Hypergossiping - Chapter 6). The main criteria for the evaluation of the algorithms were their robustness in different conditions. The simulation parameters were chosen to be able to realistically test the algorithms in environments similar to BIONETS. The results show that MMSBA provides the best coverage for the Constant Speed Mobility, but using other models, the performance both of MIOBIO and MMSBA are very close. Both of the new algorithms outperform the reference algorithms in terms of reliability and induced overhead.

Further simulations have been executed to compare the amount of network traffic generated when the nodes employ the situated adaptive opportunistic routing versus a plain epidemic routing protocol (Chapter 7). As the number of devices acting in the BIONETS network can add up to considerably high numbers, avoiding unnecessary transmissions becomes essential. In the next section a simple mode of how information dissemination algorithms can be adjusted to a non-cooperative environment is introduced, by assuming that each node has knowledge about the degree of cooperation of the other nodes (Chapter 8). We investigated the effectiveness of a simple strategy that tries to avoid the nodes that are less cooperative than a pre-specified threshold. This complements the study reported in D1.3.1 [1], where the effects of cooperation on the performance of three representative algorithms were shown. In particular, the performance of the epidemic, the two-hop relay and the binary spray-and-wait information dissemination algorithm was studied within the context of a non-cooperative environment.

Finally we have studied the real behavior of classical retransmission-based communication protocols when used in a dynamic environment (Chapter 9). This was aimed to demonstrate the limits of classical communication paradigms in a real opportunistic environment, and to validate our choice to investigate novel, encoding-based communication protocols, and to apply these results to the BIONETS simulation framework. Communications among the T-Node plane and the U-Node plane can successfully be integrated into BIONETS simulation framework by means of a traffic generator which produces data according to a realistic pattern that is described by real-world traces. The integration of such real-world traces provides the simulator with a more realistic view of the T-Node to U-Node communications.

2 Connectivity and Scalability Issues

The BIONETS network is aimed at working under the conditions presented by disconnected operations. Hence, the performance of services running on top of the BIONETS communication infrastructure will be affected by the properties of disconnected communications. This section is aimed at exploring the fundamental trade-offs related to the delay and connectivity of the BIONETS communication infrastructure. In general, it is known that networks operating under disconnected conditions can support a wide range of services which are not compromised by significant delays, such as periodical massive data exchanges, database updates, file transfers, even e-mailing and, in general, those applications where data are valid within a much longer time frame than that of topology changes. In BIONETS, in order to overcome frequent network partitions, U-nodes leverage their storage capabilities, so that the common notion of routing could be better rephrased as *store-carry-and-forward* [2]. The opportunities for pairs of nodes to communicate are ephemeral and, in this respect, communication among nodes is conveniently described by the set of node *contacts* [3, 4].

When a contact between a pair of nodes occurs, they are able to exchange a certain amount of data, namely a *message*. Contacts describe the occurrence of several cases resulting from intermittent links, ranging from the case of LEO satellite networks, where contacts have a regular pattern, to contacts generated by mobile devices, coming by chance into mutual radio range.

Due to their potential impact, mostly carried on in the related field of delay tolerant networks, most works focused on sparse mobile ad hoc networks [2, 5, 6, 4], a domain where intermittent connectivity is due to node mobility and to limited radio coverage. In such cases, traditional end-to-end communication techniques have to be enforced or replaced with some mechanisms to reconcile disrupted paths. In the case of BIONETS, the general system architecture is no longer concerned with the establishment and maintenance of end-to-end paths, but, the effect of the mobility as an engine to convey information is a major issue and deserves special focus. In particular, we are concerned with the scalability of the delay tolerant mechanisms which rules the BIONETS U-node plane, with respect to the increase of the number of nodes. In particular, we focus on the notion of *connectivity* over such networks. Our starting point is that, since connectivity is not guaranteed all the time, but over time, we can build a connectivity graph that adheres to the dynamics of the contacts occurring among nodes. When such graph gets connected, we leverage the length of the longest route in ER graphs, to obtain scaling laws in the number of nodes.

From a theoretical perspective, a reference work for mobile wireless ad hoc networks is the pioneering work of Glossgauser and Tse [7]. The results therein focused on the (broader) problem of the capacity of ad hoc networks. The main finding was that mobility can be leveraged to increase capacity. The hint was a viable direction to overcome the severe limitations in static networks capacity [8].

Along this line, later research proved that such improvement comes at the price of an inherent trade-off with the delay required to deliver a message to destination [9]. Furthermore, the authors of [10] confirmed that throughput results on the reference two-hop protocol in [7] are tight, in the sense that redundant messages cannot increase the network capacity, but, nevertheless,

redundancy is beneficial towards improving delay.

Most literature on DTNs addresses the problem of routing, where a key role is played by the knowledge of input variables such as contact times, traffic demands or memory occupation [11]. The assumptions for routing in DTNs may range from the case of perfect or partial information, when a so called *oracle* is assumed to trace input variables, to the case of routing with zero knowledge. In [2, 12], the authors describe a network architecture, where special nodes, named ferries, carry data among nodes. In [2], ferry mobility is designed to minimize the duration of the ferry tours and achieve high contact probabilities with nodes. Multiple ferries' route design with minimum delay is studied in [12]. In BIONETS, we generally assume that users' mobility cannot be modified, and for this reason we usually exclude, apriori to operate mobility control.

A different research line adopts epidemic diffusion algorithms [13, 14], also named epidemic routing or controlled flooding. A *leitmotiv* of epidemic message diffusion is, therefore how to fill the gap between pure flooding, on one hand, and the original two hop protocol in [7], on the other. The paper in [15] gives insight into the behavior of DTNs' diffusion based on the opportunistic exchange of data between nodes in radio range.

The literature reports several results of real experiments on delay tolerant networks [5, 15, 16]. In [5], the DieselNet network was deployed over a wide urban area, using buses as mobiles. Authors of [15] describe the use of human mobility to diffuse information through portable devices. In such works, authors discuss several technical issues. Special attention is given to existing routing protocols for DTNs, where the major design target is to maximize the fraction of delivered packets and to minimize the latency between source and destination. It is quite apparent that one major difficulty encountered in DTNs is the derivation of suitable robust routing algorithms. We can either leverage an *oracle* and design message delivery by node contact sequences, or we can use epidemic routing techniques instead, but message delivery ultimately relies on the underlying contact process and on a notion of connectivity over time.

Connectivity over time

Even though the common notion of a connectivity graph is well established in ad hoc network literature, defining a similar, useful notion of a connectivity graph when mobility is the major resource to ensure communications, as in the case of BIONETS architecture, is not immediate. Some proposals have been published. The authors of [3] propose the notion of a DTN multi-graph, where several edges may exist between pairs of nodes, each weighted with capacity and delay functions. Also, *space-time graphs* have been defined in [17], under perfect information of connectivity at each point in time. Space-time graphs are constructed sampling the instantaneous connectivity over a set of consecutive disjoint time intervals, where topology and link capacity do not change. Such instances are then pasted and linked by directed edges in order to construct the final instance of the space-time graph, which ultimately is an oriented graph. The construction basically corresponds to the *time-expanded graph* construction proposed in [18]. There, the effects of dynamic changes of the network edge capacities are included in a *transshipment problem* where arcs possess both a capacity and a traversal time. The corresponding maximum dynamic network

flow problem was formulated over a time-expanded graph. The works [19, 20] provide a general framework for the problem of dynamic network topology. They introduce the notion of an *evolving graph*, which is a sequence of graphs constructed from the presence schedule of all nodes and links over a set of intervals. They introduce the notion of *journey* which corresponds, over time, to the concept of a path. The authors proved that the problem of building a minimum cost-spanning tree in such domain (evolving spanning tree) is $\mathcal{NP}$ -hard [20]. Here, as in most literature, we will implicitly refer to the definition of a journey, but, since we refer to the filtering mechanisms of the BIONETS architecture, we do not assume any a priori knowledge on the dynamics of connectivity.

2.1 A connectivity graph

We consider a set of n U-nodes over an area A , each of them denoted by a unique ID, for simplicity $1, 2, 3, \dots, n$, and the notion of a contact interval denotes the period of time when two nodes are able to exchange a message. We let two nodes be able to communicate when they are within reciprocal radio range $R > 0$, and communications are bidirectional. Also, we assume that:

(a1) Contact intervals are sufficient by long to exchange all messages: this let us consider nodes' *meeting times* only, i.e., time instances at which a pair of unconnected nodes fall within reciprocal radio range;

(a2) We assume that all nodes have an infinite buffer;

In addition, the underlying assumption in the following is that the network is in the *subcritical* regime, meaning that it is not connected with high probability [21]. Furthermore, we assume that the size of each connected component is small and thus multi-hopping is not feasible.

We define as *intrameeting times* the sequence $\{Y_n\}_{n \in \mathbb{Z}}$, where $Y_n = T_n - T_{n-1}$. Finally, with standard notation, the diameter of graph G is the greatest distance between any two vertices in G . The average degree of nodes of G will be denoted d . The main definition introduced here is about a *connectivity graph*, in the following simply connectivity graph. Given a the *marked point process* associated with our system, $\{Z_n\}_n$, the connectivity graph $G^{(t_0, D)} = (V, E)$ is a graph where the set of vertices is $V = \{1, 2, \dots, n\}$ and the set of edges E

$$(i, j) \in E \quad \text{if and only if} \quad \exists T_n \text{ such that: } t_0 \leq T_n \leq t_0 + D, \sigma_n = (i, j). \quad (1)$$

In the following, we will denote parameter D as the *window length*.

The above definition also describes the simple procedure to construct the connectivity graph. We start with an empty graph with no edges and n nodes at time t_0 , and we parse the time interval up to time $t_0 + D$. Each time we register a contact between a pair of nodes (i, j) , if such a pair of nodes meet for the first time, then we add the corresponding edge (i, j) to the set of edges of the connectivity graph. In Fig. 1, we reported an example of the connectivity graph which corresponds to the mobility pattern of 5 nodes. The connectivity graph is built over time: each time a new pair of nodes meet, an edge is drawn between them. The mobility pattern, in particular, is captured by the sequence of marks corresponding to intrameetings, stamped along the time axis. We denote two features of the connectivity graph that are clarified from the example. First, replicated edges

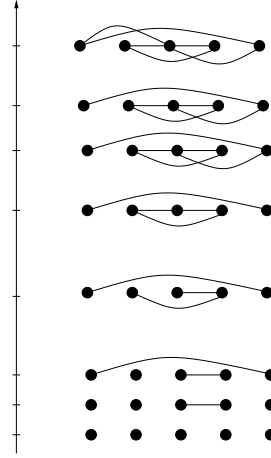


Figure 1: Example of the construction of a connectivity graph. A realization for the sequence of the intrameeting times is reported along the vertical time axis at left, marks are reported on the right of the axis.

are counted only once; we observe that edge $(3, 2)$ is counted once, even if the corresponding mark appears twice. The connectivity graph collects, to some extent, the *presence* information on the contact process, meaning that if an edge exists, then at least one contact between two nodes appeared in the given interval. Second, the information on the *order* of the sequence of contacts between nodes is not preserved. In practice, even if nodes 2 and 5 are connected in the resulting graph, this does not guarantee that a message can propagate along such a path: for example, node 5 could exchange a message and relay it to node 2 using node 3 as an intermediate relay, but the converse is not possible. Nevertheless, the cumulative information on the contact process retained by the connectivity graph let us obtain scaling laws for the delay in DTN networks¹. In particular, we will refer to the class of mobility patterns, named the Marks-Memoryless (M^2), introduced in [22, 1]; for the ease of reading, the main features of the M^2 model are recalled here. The M^2 class collects all the mobility patterns where the mobility pattern is such that i) marks $\{\sigma_n\}$ are i.i.d, ii) *intrameeting times* $\{Y_n\}_{n \in \mathbb{Z}}$ are nonnegative i.i.d. The statistical tests performed in [22], proved that the M^2 shows a very good statistical fit with mobility models used in ad hoc networking, such as Random Waypoint Mobility (RWM), Random Direction Model (RDM), etc. [23].

2.2 Diffusion and Bounds

As the connectivity graph carries the cumulative information on the meetings occurring up to a given instant, its evolution depends on the window length D considered. Furthermore, under the hypothesis that $\pi(i, j) > 0$ for $\forall i, j$, it is trivial that the connectivity graph will converge to the complete graph as $D \rightarrow \infty$. Therefore, edge (i, j) will belong to the connectivity graph with a

¹We note that the connectivity graph can be defined for any system ruled by a sequence of intrameetings, but, to simplify our analysis, we have limited our study to encounters that can be modeled through a particular class of mobility patterns.

probability which in general is a continuous, non-decreasing function of the window size D .

We introduce a further uniformity assumption, meaning that all nodes move according to the same mobility model. Hence, by symmetry, the marks will have a uniform distribution, which means $\pi(i, j) = 1/C$, where $C = \binom{n}{2}$ is the maximum number of possible edges. If this is the case, the growth of the connectivity graph described by (1), will be performed adding incrementally edges at each intrameeting, but with no preferential attachment. Hence, we are basically excluding the occurrence of power laws [24], which might appear in the case of non-uniform marks distribution. Thus, $p_{ij}(D) = p$ is independent of the considered edge. This means that the connectivity graph will be an Erdős-Renyi random graph [25].

In the following we will use two features of ER graphs. The first property states that connectivity occurs abruptly above a corresponding threshold value for $p(n)$. The scaling of $p(n)$ for ER graphs is such that almost all ER graphs will be connected when the average degree of a node, $p(n-1)$, scales faster than $\log(n)$, and is disconnected below such threshold. Second, we will employ a result on the diameter of ER random graphs, stating that above the threshold, the diameter of random graphs is almost surely *concentrated* on a few values around $\frac{\log n}{\log np}$ [26]. For general properties of most graphs see [27, 25].

It is possible to leverage general properties of ER graphs to characterize how the typical interval of observation should scale in order to make the connectivity graph connected. In particular, we want to determine the corresponding scaling law of the window length, at the increase of the number of nodes. It turns out that the time needed in order for all the nodes to be connected in the connectivity graph grows as a mild superlinear function of n . In what follows, we will briefly refer to D^* as the *critical window (length)*.

2.2.1 Critical window and delay bound

In [28] the number of nodes and the critical window D^* are related through the average number of edges of the connectivity graph. It turns out that D^* can be expressed directly as a function of n , as shown in the next sections. Then, the following result holds:²

Theorem 1 *For n mobile nodes, satisfying a MM^2 mobility model, it holds $D^* = \Theta\left(\frac{n \log(n)}{\lambda}\right)$.*

The proof of the above statement is derived in [28]. There, direct insight into the dynamics of connectivity are provided using a fluid approximation, under the assumption that $n \gg 1$.

Now we assume that a message is generated by node i and intended for node j : since the system is subcritically connected, with unitary probability, the message cannot be delivered using *any* multihopping technique, the reason being simply that almost surely a path between any two nodes does not exist. Nevertheless, using opportunistic forwarding it is possible to deliver the message, storing it at intermediate nodes. We note that under a Marks Memoryless model, using either an epidemic diffusion mechanism [13, 14] or the two-hop routing mechanism [7], the

²Given two functions $f(n)$ and $g(n)$, we use the following notation: (i) $f(n) = o(g(n))$ if $\frac{f(n)}{g(n)} \rightarrow 0$ as $n \rightarrow +\infty$, (ii) $f(n) = O(g(n))$ if $\frac{f(n)}{g(n)}$ is upper bounded for n large enough, (iii) $f(n) = \omega(g(n))$ if $g(n) = o(f(n))$, (iv) $f(n) = \Omega(g(n))$ if $g(n) = O(f(n))$ and (v) $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.



Figure 2: Degree distribution of connectivity graphs and comparison with the ER model, RWP mobility model; window length $D = 10, 100, 200, 300, 450, 600$ s.

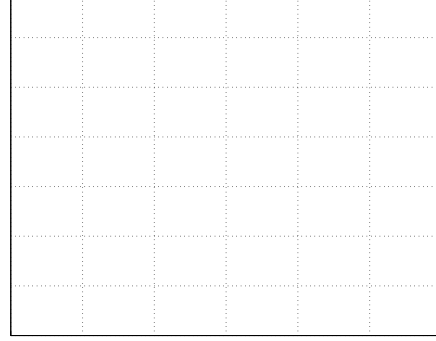


Figure 3: Clustering coefficient versus window length, RWP mobility model.

message will be delivered to the destination node with unitary probability over an infinite time interval. As previously observed, an important issue is the mean delay for a message originating at node i to reach node j , which we denote $\mathcal{D}_{ij}$. Finally, we find the following scaling law for the message delay, whose proof is deferred to the Appendix.

Theorem 2 *For n mobile nodes, satisfying a MM^2 mobility model, it holds $\mathcal{D}_{ij} = O\left(\frac{n \log^2(n)}{\lambda \log(\log(n))}\right)$.*

As we described before, the existence of a path between two nodes in the connectivity graph does not guarantee that a message can be delivered between the end nodes. Nevertheless, the scaling law of D^* gives insight on how to dimension certain routing parameters. One of these parameters, which is quite critical in BIONETS epidemic routing, is the duration of timeouts after which a message can be safely discarded. In the case of epidemic routing, in fact, this becomes a major concern, due to the replication of multiple copies of the same message. Under disconnected operations, in practice, a common paradigm is the use of k -relaying protocols [15], where every node replicates a copy of a message k times to nodes not already receiving the message. The indications from Thm. 1 is that message timeouts should scale as $\Omega(n \log(n)/\lambda)$. Also, if messages are acknowledged at the receiver side, the result of Thm. 2 suggests that the timeout scaling should be $O\left(\frac{n \log^2(n)}{\lambda \log(\log(n))}\right)$. We notice that the result in Thm. 2 also provides a loose lower bound on the storage requirements of U-nodes in BIONETS. If the message generation rate at each node is $0 < \mu < \infty$, a straightforward application of Little's law suggests that the number of messages stored will increase at least as $\mu \mathcal{D}_{ij}$. Thus, neglecting the storage for relaying, memory requirements are $\Omega\left(\frac{n \log^2(n)}{\lambda \log(\log(n))}\right)$.

2.3 Numerical Results

As we described before, one main rationale underlying our model is that, under uniformity assumptions on mobility, the connectivity graph matches an ER graph. We validated our model

comparing the typical parameters [27] of the connectivity graph with the corresponding values of a ER random graph. The tests were performed as follows: Given a trace of the contact process, and a value of D , we generated the connectivity graphs over consecutive disjoint intervals. For each realization of a connectivity graph, we calculated the corresponding ER graph parameter $p = d/(n - 1)$. Then we averaged and compared the *degree distribution*, the *clustering coefficient* and the graph *diameter*. The clustering coefficient of a node i is the ratio $\mathcal{C}\ell_i = T_i / \binom{k_i}{2}$, where T_i is the number of edges among the k_i neighbors of node i , and the size of a complete graph with the same number of nodes; the clustering coefficient of graph G is the average of $\mathcal{C}\ell_i$ over the set of nodes. The clustering coefficient describes the probability of a node to belong to a complete cluster of nodes [27]. For a ER random graph of parameter p , $\mathcal{C}\ell = p$.

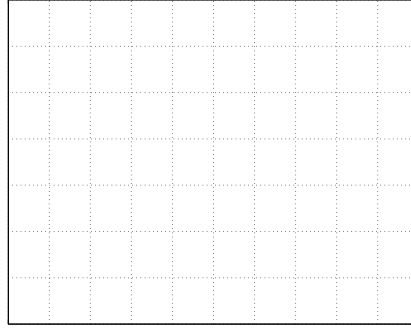


Figure 4: Diameter versus window length, RWP mobility model.

Synthetic mobility traces. We first considered the contact traces recorded in a simulation of a set of $N = 60$ mobile nodes moving according to a Random Waypoint mobility model, over a square area of 10000 m^2 with a speed of 5 m/s and zero pause intervals; the initial set-up of simulations adhered to the stationarity conditions indicated in [29]. The simulations were performed using the Omnet++ simulation platform [30].

The first quantity considered is the probability mass distribution of nodes' degree. As depicted in Fig. 2, connectivity graphs match the binomial degree distribution of a ER graph. Nevertheless, certain structural properties of graphs are not caught by the plain degree distribution. Thus, we further tested the clustering coefficient and, as depicted in Fig. 3, we found a very good match. Notice that for small values of the window length, the clustering coefficient is over-estimated: this happens because a large fraction of isolated nodes exists, thus lowering the average value. Finally, we tested the behavior of the diameter of connectivity graphs at the increase of the window length, reporting also the reference trend for the ER model, when samples are generated with the same parameter p .³ As expected, the connectivity graphs show a decreasing diameter at the increase of the window length. We notice that the diameter of the connectivity graphs resemble closely the same trend shown by the ER graphs. In the case of the RWP mobility, the very good match with our model is not surprising, since the underlying mobility process provides a fairly uniform distribution in the marks.

³We excluded from the average instances with infinite diameter.

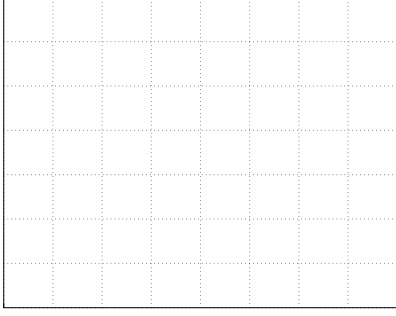


Figure 5: Degree distribution of connectivity graphs and comparison with the ER model, DieselNet trace 3022005; window length $D = 1000, 30000$ s.

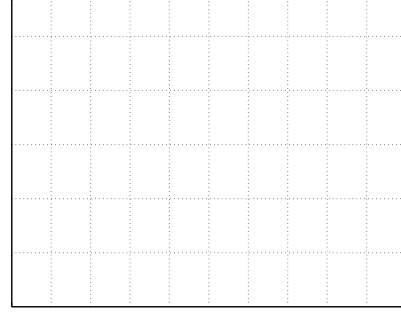


Figure 6: Clustering coefficient versus window length, DieselNet trace 3022005.

Real-world traces We performed further tests on the contact traces generated by the DieselNet net-

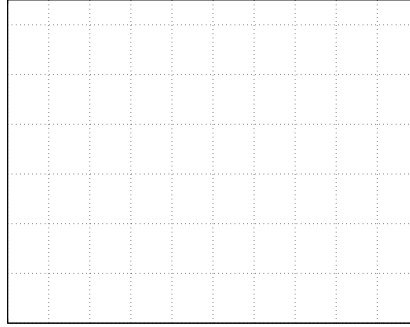


Figure 7: Diameter versus window length, DieselNet trace 3022005.

works, and available online [31, 5]: Such contact traces were collected on a daily bases, and the core network is represented by 20 devices, using a IEEE802.11 interface and mounted on buses. A preliminary analysis showed that the mark distribution, in this case, is *not* uniform, and, moreover, certain contacts do not appear in the trace, meaning that, as expected, certain nodes do not meet at all (this is quite intuitive since, usually, not all bus routes are supposed to intersect). We note that, in the case of the DieselNet, the contact traces are clearly *not stationary*, and to this extent, p represents only a benchmark for our model; one major consequence is that p is not guaranteed to be monotonically non decreasing, depending both on the window length and the initial epoch of observation t_0 .

In order to have a significant number of contact events, we did not considered nodes which generated less than 20 contacts per day. Thus, the network is restricted to a subnetwork of 17 nodes, for a run of 76686 s. In the overall trace, the maximum number of possible contact pairs was $C = 136$, but 56 pairs were not present. The mean intrameeting time is 156.629 s with a maximum intrameeting time of 2262 s. The degree distribution shows that the connectivity graph has

a heavier tail for the real-world trace, meaning that there exist a few nodes which connect more than others (preferential connectivity), for example, as reported in Fig. 5, for $D = 30000$ s there is a larger frequency of nodes of degree 10 and 11 than expected, and, meanwhile, a non-negligible number of nodes which experienced only one contact.

The tests on the clustering, see Fig. 6, confirm that, as we expected, real-world traces have clustering coefficients higher than in the corresponding ER model, as it appears at higher values of the window length. Furthermore, the diameter is higher than in the case of the ER model, as shown in Fig. 7. This confirms the results on clustering, indicating a certain *locality* in the contact sequence between nodes.

3 Network Design and Optimization: Clustered Infrastructure

In this section, we consider a clustered infrastructure. Our assumption that T-nodes are grouped in clusters. Each cluster is centered around a single U-node, which collects all the information from the T-nodes that surround it.

The performance of the clustered system with interference from neighboring clusters is analyzed. A useful framework is provided in order to determine the U-node coverage for wireless frequency-selective fading channels with very dense networks (in the number of T-nodes per meter) considering different receiver structures at the U-node. Using asymptotic arguments, analytical expressions of the spectral efficiency are obtained and provide a simple expression of the network capacity based on only a few meaningful parameters.

T-nodes are assumed to always have information to transmit. In order for the U-node to be able to receive information from all T-nodes in the cluster at the same time, our assumption is that spreading codes are used. Spreading codes are a typical feature of the non-orthogonal multiple-access physical layer protocol CDMA (code division multiple access) [32], which enables multiple transmitters to transmit simultaneously and on the same frequency band.

CDMA is a multiple access physical protocol that has been extensively studied and published. When considering that all spreading codes are known at the receiver, the performances of multi-user systems has been thoroughly investigated by the information theory community, see for example [33, 34, 35, 36]. Extensions to a more involved attenuation model can be found in [37].

However, when there are interferers with unknown codes, the communications are perturbed both by transmitters with known spreading sequences as well as unknown spreading sequences. This is akin to multi-cell networks. In order to obtain tractable expressions, studies of multi-cell networks were restricted to the study of a few interfering cells with simple interference models [38, 39]. None took the cumulative effect from all interfering cells with realistic path loss explicitly into account. A new approach was proposed in the downlink case in [40, 41], as well as in the uplink case in [42]. The latter contribution was restricted to flat fading. However, wireless transmissions generally suffer from the effect of multiple paths, thus becoming frequency-selective.

We analyze a clustered setting, in the case of frequency selective fading, for two types of receiver structures: The matched filter and Optimum filter. The framework introduced in [41, 42] is used to model the environment given a uniformly distributed density (number of T-nodes per meter) of the T-nodes as well as certain characteristics of the fading channel. The clusters are spatial groupings of T-nodes, centered around a receiving U-node. In order to obtain analytical expressions, we consider the system in an asymptotic setting, letting both the number of T-nodes and the spreading factor trend to infinity with a fixed ratio. We employ random matrix theory tools [43] to analyze the system in this limit. Random matrix theory is a field of mathematical physics that has been recently applied to wireless communications to analyze various measures of interest such as capacity or Signal to Interference plus Noise Ratio (SINR). Interestingly, it enables to singling the main parameters of interest that determine the performance in numerous models of communication systems with more or less involved models of attenuation. This process is

detailed in a recent survey on random matrix theory which focuses on wireless communications [43]. In addition, these asymptotic results provide good approximations for the practical finite size case (spreading length N of 256), as shown by simulations.

3.1 Model and Notations

We focus our analysis on a one dimensional (1-D) network. This scenario represents, for example, the case of the deployment of T-nodes along a motorway (U-nodes, i.e., cars move along the motorway). The U-nodes are supposedly equidistant with inter-U-node distance a . Clusters are determined spatially and regroup T-nodes that are closer to the central U-node.

An infinite length $((2M + 1)a$ with $M \rightarrow \infty$) U-node deployment is considered. The spreading length N is fixed and is independent of the number of T-nodes. T-nodes are uniformly distributed along the line with density d . The number of T-nodes per cluster is $K = da$ and the load of each cluster is $\frac{K}{N} = \alpha a$.

Without lack of generality, we focus on a cluster with U-node located at the origin. This cluster gathers T-nodes numbered from 1 to K . We consider the case of a multipath channel.

We would like to quantify the number of bits/s/Hz the system is able to deliver to all the T-nodes given a certain inter-cluster distance a . The T-nodes are assumed to employ Gaussian codebooks. Due to invariance by translation, the spectral efficiency per cell is the same for all cells. Since the network capacity is infinite, the measure of performance in this case is the number of bits per second per hertz per meter (bits/s/Hz/m) the system is able to deliver. For linear detectors such as Matched Filter or MMSE Filter, an SINR can be defined, and the spectral efficiency is expressed:

$$\gamma = \frac{1}{FNa} \sum_{k=1}^K \log_2(1 + \text{SINR}_k). \quad (2)$$

In general, the SINR is a random variable, and we will consider the mean spectral efficiency, averaged over the fading distribution.

3.2 Matched Filter

We consider both the case when U-nodes of each cluster operate independently, as well as the case when all U-nodes are linked to a wired infrastructure and exchange information, in order to get a better estimation of the interference and to perform joint processing.

3.2.1 Infinite Cluster Network

Proposition 1 *As $N, K \rightarrow \infty$ with $K/N \rightarrow \alpha a$, the SINR of T-node k at the output of the matched filter is given by*

$$\text{SINR}_k = \beta^{\text{MF}}(x_k)$$

where $\beta^{\text{MF}} : [-a/2, a/2] \rightarrow \mathbb{R}$ is given by

$$\beta^{\text{MF}}(x) = P(x) \cdot \frac{\left(\int_0^1 |h(f, x)|^2 df \right)^2}{\sigma^2 \int_0^1 |h(f, x)|^2 df + \int_{-\frac{a}{2}}^{+\infty} \int_0^1 P(y) |h(f, y)|^2 |h(f, x)|^2 df dy}. \quad (3)$$

3.2.2 Joint Cluster Processing

The matched filter gives the same SINR results for each T-node, but is simply applied to a much greater (infinite) number of T-nodes.

3.3 Optimum receiver

As in the previous section, we consider both cases of independent and joint processing of the signals received by U-nodes.

3.3.1 Infinite Cluster Network

Let $\mathbf{R}^+ = \sum_{l=1}^{\infty} (\mathbf{H}_l \sqrt{\mathbf{P}_l} \odot \mathbf{W}_l) (\mathbf{H}_l \sqrt{\mathbf{P}_l} \odot \mathbf{W}_l)^H$ and $\mathbf{R}_0^+ = \mathbf{R}_0 + \mathbf{R}^+$.

Proposition 2 When $N \rightarrow \infty$ and $\frac{d}{N} \rightarrow \alpha$, the spectral efficiency with i.i.d. random spreading and optimum filter is:

$$\gamma = \frac{1}{a \ln(2)} \int_{+\infty}^{\sigma^2} \left(m^{\mathbf{R}_0^+}(-z) - m^{\mathbf{R}^+}(-z) \right) dz \quad (4)$$

where $m^{\mathbf{R}_0^+}(z)$ and $m^{\mathbf{R}^+}(z)$ are the Stieltjes transforms of the empirical distribution functions of the eigenvalues of $\mathbf{R}_0^+$ and $\mathbf{R}^+$ given by

$$\begin{aligned} m^{\mathbf{R}_0^+}(z) &= \int_0^1 t(f, z) df, \\ m^{\mathbf{R}^+}(z) &= \int_0^1 v(f, z) df \end{aligned}$$

and $t(f, z)$, $v(f, z)$ satisfy the fixed point equations:

$$\begin{aligned} t(f, z) &= \frac{1}{\int_{-a/2}^{+\infty} \frac{P(x) |h(f, x)|^2 dx}{1 + \int_0^1 P(x) |h(f', x)|^2 t(f', z) df'} - z}, \\ v(f, z) &= \frac{1}{\int_{a/2}^{+\infty} \frac{P(x) |h(f, x)|^2 dx}{1 + \int_0^1 P(x) |h(f', x)|^2 v(f', z) df'} - z}. \end{aligned}$$

3.3.2 Joint Cluster Processing

Proposition 3 When $N \rightarrow \infty$ and $\frac{d}{N} \rightarrow \alpha$, the spectral efficiency with i.i.d. random spreading, optimum filter and joint cluster processing is:

$$\gamma = \frac{1}{a \ln(2)} \int_{+\infty}^{\sigma^2} \left(m^{\mathbf{R}_0^+}(-z) - \frac{1}{z} \right) dz. \quad (5)$$

3.4 Conclusion

Using asymptotic arguments, an explicit expression of the spectral efficiency of an infinite deployment of T-nodes grouped in clusters around a central U-node has been derived considering realistic path loss and fading models. We have shown in particular the potential gain of optimum

processing with respect to various receivers. The results are especially useful for the deployment of U-nodes for a given target T-node rate. Note that although the model under consideration applies to 1-D networks, it is straightforward to extend the analysis to 2-D networks (in the case of a regular pattern).

4 Network Design and Optimization: Ad-Hoc Network Infrastructure

We have studied the capacity scaling of a system where a source U-node communicates to a destination U-node with the help of several scattering T-nodes. Capacity expressions accounting for physical characteristics of the environment (topology, frequency band...) are provided and an asymptotic analysis is performed for the increasing size of the dense scattering environment. The capacity is shown to reach a saturation level in the asymptotic regime, suggesting a capacity-delay trade-off. Moreover the saturation point depends on the positioning of T-nodes, in particular in wide band systems where topology impacts capacity in terms of both pathloss and delays. Waiting very long for retransmissions from an infinite number of T-nodes is not practicable and a few well located T-nodes around the source and destination lead to better performances than more T-nodes uniformly distributed on a square area between the source and destination.

4.1 Introduction

The issue of how the performance of virtual MIMO systems scale with the number of nodes or size of the network is a crucial issue, and since the pioneering work by Gupta and Kumar [44], several recent works [45], [46] considered capacity scaling laws in large narrow-band networks. Nevertheless, no previous work focused on the scaling of capacity in a dense wideband – thus with high resolution in time and space – networks of dumb scatterers taking into account topology not only in terms of pathloss but also of multi-path. Moreover, the environmental impact on the scaling laws, through reflections, diffraction effects, etc. has never been studied in details.

We analyze a system where a dense topology of T-nodes helps a source U-node to communicate with a destination U-node and we look at the network from a physical propagation point of view: relays are modeled as dumb omnidirectional T-nodes, i.e. passive nodes without engineering capabilities that simply scatter the incident electromagnetic wave emanating from a source antenna. Capacity expressions are given as a function of the physical characteristics of the network, including topology and signal band. We study how capacity scales when the size of the network increases, as well as the point at which asymptotic regime is reached depending on the nodes positioning. Interestingly the asymptotic analysis, and simulations in wideband systems, show that after a certain point, capacity saturation occurs. This is due to the fact that signal contributions coming from peripheral nodes significantly distant from source and destination do not lead to significantly increased performance, suggesting a capacity/delay trade-off, which may not be the case when adding MIMO capability.

4.2 System Model

We focus on a geometrical network model by considering a finite square grid of T-nodes, spaced by $\lambda_s/2 = c/(2W)$ vertically and horizontally, where W is the sampling rate. Source S and Destination D are located on vertices of the grid separated by a distance $d_0 = m\lambda_s/2$. The carrier wavelength will be denoted $\lambda_c = c/f_c$.

4.2.1 T-nodes Clustering

In a wide band system, resolution in time and space are high and propagation delays cannot be neglected as in narrow band systems. Indeed λ_s may be small compared to the distance between communicating terminals, therefore a signal transmitted during a given time-slot will not be received in the same time-slot but later. To take into account propagation delays, T-nodes are grouped in N_{clus} clusters, depending on the time-slot of reception of the scattered wave.

If we sample at $W = c/\lambda_s$, we can discriminate durations greater than the time $T = \lambda_s/c$ needed by the wave to propagate along λ_s meters. Thus a cluster gathers T-nodes whose scattered waves are received by D during the same interval T .

The shortest distance to go from source to destination is the straight path without reflection of length d_0 and propagation duration $\tau_0 = d_0/c = (m/2)T$. Cluster Δ_0 contains the straight path from source to destination, as well as the reflected paths received during the first time-slot $TS_0 = [\tau_0, \tau_0 + T[$. Those paths correspond to waves propagating on a total distance, sum of the distances from S to the T-node and from the T-node to D , $d_r^{(s)} + d_r^{(d)} \in [d_0, d_0 + \lambda_s[$. For $i \geq 1$, Cluster Δ_i contains only scattered waves, such that $(d_r^{(s)} + d_r^{(d)}) \in [d_0 + i\lambda_s, d_0 + (i+1)\lambda_s[$. Those waves are received during the i^{th} time-slot defined by $TS_i = [\tau_0 + iT, \tau_0 + (i+1)T[$. For a T-node k in cluster i , the propagation delay of the scattered wave from S to D via this T-node $R_{k,i}$ will be denoted $\tau_{k,i} \in TS_i$. This duration consists of three parts: $\tau_{k,i} = \tau_0 + iT + \tau'_{k,i}$ with $\tau'_{k,i} \in [0, T[$.

4.2.2 Power Attenuation

To model the attenuation of propagating waves, the far field propagation model holding for distances $d \gg \frac{\lambda_c}{2\pi}$ will be considered. Since $f_c \geq W/2 \Leftrightarrow \lambda_s \geq \lambda_c/2$, the far field condition is fulfilled for distances $d \geq \lambda_s/2$. The power received by D depends on the path taken by the signal to reach D : one-hop direct path from S to D without reflection or two-hop-path from S to D via one T-node. Multiple reflections before reaching D are not taken into account.

The direct received power P_d and the reflected power P_r received at D after one reflection are [47, 48]:

$$\begin{aligned} P_d &= \frac{\lambda_c^2}{(4\pi)^2 d_0^2} P_{rad} = K_1^2 \frac{\lambda_c^2}{d_0^2} P_{rad} \\ P_r &= \frac{\lambda_c^2 s}{(4\pi)^3 (d_r^{(s)} d_r^{(d)})^2} P_{rad} = K_2^2 \frac{\lambda_c^2}{(d_r^{(s)} d_r^{(d)})^2} P_{rad} \end{aligned} \quad (6)$$

with P_{rad} the power transmitted by the source antenna, s the radar cross section, and the constants $K_1 = 1/(4\pi)$ and $K_2 = \sqrt{s/4\pi} K_1 = \sqrt{s/(4\pi)^3}$.

4.3 Analysis

4.3.1 Signals Expressions in Time Domain

Source S produces a sequence $s(t) = \sum_{n=0}^{N-1} s_n \delta(t - nT)$ of N complex symbols with unit energy $\varepsilon = E[|s_n|^2] = P_{rad}$ at rate $W = 1/T$ and transmits a linearly-modulated signal with complex envelope $x(t) = s(t) * g(t) = \sum_{n=0}^{N-1} s_n g(t - nT)$, where $g(t)$ is the pulse shaping filter, satisfying the Nyquist criterion $\int g(t)g^*(t - kT)dt = \delta_{0,k}$. We assume hereafter that $g(t)$ is the rectangle function of amplitude $1/\sqrt{T}$ over $[0, T]$. The received signal at D is the superposition of the signal coming directly from S and the signals scattered once:

$$y(t) = \sum_{i=0}^{N_{clus}-1} \sum_{k \in \Delta_i} a_{k,i} x(t - \tau_{k,i}) + n'(t) \quad (7)$$

where $n'(t)$ is AGWN and coefficients $a_{k,i}$ are given by (6):

$$a_{k,i} = \begin{cases} K_1 \frac{\lambda_c}{d_0} e^{j\varphi_{0,0}} & \text{for } (k,i) = (0,0) \text{ direct path} \\ K_2 \frac{\lambda_c}{d_{k,i}^{(s)} d_{k,i}^{(d)}} e^{j\varphi_{k,i}} & \text{for } (k,i) \neq (0,0) \text{ reflected path} \end{cases}$$

$\varphi_{k,i} \in [0, 2\pi[$ are phase shifts due to propagation and reflections. Phases can be modeled as independent random variables provided nodes are sufficiently spaced.

After matched-filtering and sampling at rate $W = 1/T$, the received signal becomes:

$$r_l = r(lT) = \sum_{n=0}^{N-1} s_n h(lT - nT - \tau_0) + n(lT) = \sum_{n=0}^{N_{clus}} h_n s_{l-n-m/2} + n_l \quad (8)$$

with the T-nodes network equivalent channel defined by $h(t) = \sum_{i=0}^{N_{clus}-1} \sum_{k \in \Delta_i} a_{k,i} g(t - \tau_{k,i} + \tau_0) * g^*(-t)$. The sum in (8) is finite and contains only $N_{clus} + 1$ non-null terms, since the channel impulse response can be shown to have finite length $N_{clus} + 1$. Because of the minimum propagation delay $\tau_0 = mT/2$, the $m/2$ first samples $r_0 = \dots r_{m/2-1} = 0$ are null.

Under matrix notation, the system can be reduced to the equation $\mathbf{R}_N = \mathbf{H}_N \mathbf{S}_N + \mathbf{N}_N$, where $\mathbf{S}_N = [s_0, \dots, s_{N-1}]^T$, $\mathbf{N}_N = [n_0, \dots, n_{N-1}]^T$, $\mathbf{R}_N = [r_{m/2}, \dots, r_{N+m/2-1}]^T$ are columns of size N and the channel is represented by $N \times N$ lower triangular banded Toeplitz matrix whose first column is $[h_0, \dots, h_{N_{clus}}, 0, \dots, 0]^T$.

4.3.2 Asymptotic analysis for Capacity expression

Using Toeplitz and circulant matrices properties [49], we can show that the network capacity is:

$$C = \lim_{N \rightarrow +\infty} \frac{W}{N} \log_2 \det \left(\mathbf{I}_N + \frac{\varepsilon}{W\sigma^2} \mathbf{H}_N \mathbf{H}_N^H \right) = \lim_{N \rightarrow +\infty} \frac{W}{N} \sum_{k=0}^{N-1} \log_2 \left(1 + \frac{\varepsilon}{W\sigma^2} |\lambda_{k,N}|^2 \right) \quad (9)$$

where $\{\lambda_{k,N}\}_{k \in [0, N-1]}$ are given by the N-point DFT of the first column of $\mathbf{H}_N$.

Expressing the average capacity in terms of the network physical characteristics is intricate. Nevertheless Jensen's inequality provides for an upperbound (10) to the average capacity $\bar{C} = \mathbb{E}_h [C]$ in the function of physical characteristics of the network. Besides at low SNR $\rho = \frac{\varepsilon}{\sigma^2}$, using the Taylor expansion of the log around zero at order 1, $\bar{C}$ can be written as in (11). They illustrate the SNR gain due to T-nodes in the function of the topology and the system band. We would like to point out that those formulas are valid for any topology and are not specific to the grid model.

$$\bar{C} \leq W \log \left(1 + \frac{\rho}{W} \left(K_1^2 \frac{\lambda_c^2}{d_0^2} + K_2^2 \frac{\lambda_c^2}{\lambda_s^2} \sum_{i=0}^{N_{clus}-1} \sum_{k \in \Delta_i} \frac{((i+1)\lambda_s + d_0 - d_{k,i}^{(s)} - d_{k,i}^{(d)})^2 + (i\lambda_s + d_0 - d_{k,i}^{(s)} - d_{k,i}^{(d)})^2}{(d_{k,i}^{(s)} d_{k,i}^{(d)})^2} \right) \right) \quad (10)$$

$$\text{At low SNR: } \bar{C} \approx \frac{\rho}{\ln 2} \left(K_1^2 \frac{\lambda_c^2}{d_0^2} + K_2^2 \frac{\lambda_c^2}{\lambda_s^2} \sum_{i=0}^{N_{clus}-1} \sum_{k \in \Delta_i} \frac{((i+1)\lambda_s + d_0 - d_{k,i}^{(s)} - d_{k,i}^{(d)})^2 + (i\lambda_s + d_0 - d_{k,i}^{(s)} - d_{k,i}^{(d)})^2}{(d_{k,i}^{(s)} d_{k,i}^{(d)})^2} \right) \quad (11)$$

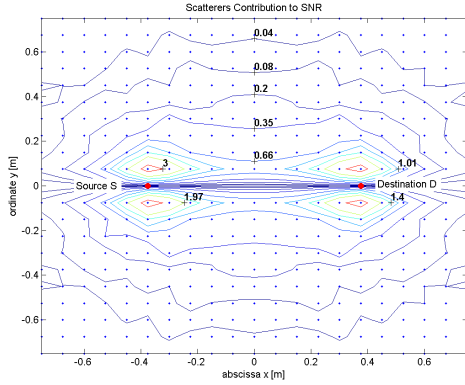


Figure 8: Contribution to SNR of T-nodes vs. their position

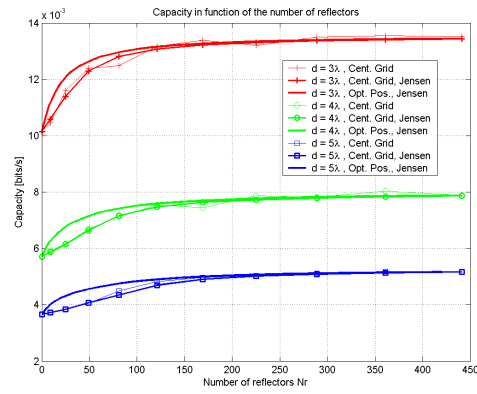


Figure 9: Capacity in function of the number of T-nodes

4.4 Simulations and Results

Simulation in a wide band system with $W = 2GHz$ and a carrier $f_c = 2GHz$, at SNR $\rho = 10dB$ allow for the analysis of the growth of average capacity when the number of T-nodes increases. We take into consideration that T-nodes are located on the vertices of a square grid covering an area of $21\lambda_s/2 \times 21\lambda_s/2$ and that the line (SD) does not contain any T-node, since when three nodes are aligned, one link among the links S-R, S-D, R-D is blocked.

As the number of T-nodes increases, capacity saturation occurs, since signals coming from peripheral nodes distant from source and destination lead to small contributions, suggesting a capacity/delay trade-off. With respect to the case without T-nodes, the level of saturation corresponds to a 30% increase in capacity at $d_0 = 3\lambda_s$ and 40% at $d_0 = 5\lambda_s$.

Simulations also show that the contribution to SNR of each T-node depends on its position. T-nodes, positions affect the capacity not only in terms of path loss but also of delay, leading to a notable difference between performances in the centered-grid case and the optimal positioning case. A few well-located T-nodes, close to source or destination according to fig. (8), lead to better performance than a great number of T-nodes uniformly distributed between S and D.

4.5 Conclusion

In this contribution, the scaling behavior of capacity in a dense scattering network is analyzed from a physical point of view, taking into account characteristics such as topology and trans-

mission band. Asymptotic analysis shows that capacity saturates when the size of the network increases and that topology affects the saturation point, in particular in wide band systems where the impact of topology on capacity is not only a matter of pathloss but also of delays that cannot be neglected. Capacity saturation suggests a capacity/delay trade-off : it is not practicable waiting for infinite retransmissions, and a few well-located T-nodes around source and destination lead to better performance than more T-nodes uniformly distributed on a square area centered between source and destination.

5 Scalability under energy and bandwidth constraints

Using game-theoretic tools, a useful framework is provided in order to determine the optimal power allocation when several T-nodes transmit to a single U-node, knowing only their own channel (while perfect channel state information is assumed at the U-node). We consider the realistic case of frequency-selective channels. This scenario illustrates the case of decentralized schemes, where limited information on the network is available at the T-nodes. Various receivers are considered, namely the Matched filter, the MMSE filter and the optimum filter. The goal is to derive simple expressions for the non-cooperative Nash equilibrium as the number of T-nodes becomes large and the spreading length increases. To that end two asymptotic methodologies are combined. The first is asymptotic random matrix theory which allows us to obtain explicit expressions of the impact of all other T-nodes on any given tagged T-node. The second is the theory of non-atomic games which computes good approximations of the Nash equilibrium as the number of T-nodes grows.

5.1 Introduction

Resource allocation is of major interest in the context of multi-user systems. In the uplink multi-user systems, it is important for T-nodes to transmit with enough power to achieve their requested quality of service, but also to minimize the amount of interference caused to other T-nodes. Thus, an efficient power allocation mechanism allows the prevention of an excessive consumption of the limited resources of the T-nodes.

In this context, a natural framework is game theory, which studies competition (as well as cooperation) between independent actors. Tools of game theory have already been frequently used as a central framework for modeling competition and cooperation in networking - see for example [50] and references therein. Building on the framework of [51], a game-theoretic approach was introduced in [52, 53].

Game theory can be used to treat the case of any number of players. However, as the size of the system increases, the number of parameters increases drastically and it is difficult to gain insight on the expressions obtained.

In order to obtain expressions depending only on few parameters, we consider the system in an asymptotic setting, letting both the number of T-nodes and the spreading factor tend to infinity with a fixed ratio. We use tools of random matrix theory [43] to analyze the system in this limit.

In the asymptotic regime, the non-cooperative game becomes a non-atomic one, in which the impact (through interference) of any single T-node on the performance of other T-nodes is negligible. In the networking game context, the related solution concept is often called Wardrop equilibrium [54]; it is often much easier to compute than the original Nash equilibrium [50]. In this paper, we derive the non-atomic equilibrium, which generally corresponds to a non-uniform PA for the T-nodes.

In [55], defining the utility as advised in [56] as the ratio of the throughput to the transmission power, the authors obtain results of existence and unicity of a Nash equilibrium for a CDMA

system. They extend this work to the case of multiple carriers in [57]. In particular, it is shown that T-nodes will select and only transmit over their best carrier. As far as the attenuation is concerned, the consideration is restricted to flat fading in [55] and in [57] (each carrier being flat fading in the latter). However, wireless transmissions generally suffer from the effect of multiple paths, thus becoming frequency-selective.

This work is an extension of [55] in the case of frequency-selective fading, in the framework of multi-user systems. We do not consider multiple carriers, as in [57], and the results are very different to those obtained in that work. The extension is not trivial and involves advanced results on random matrices with non-equal variances due to Girko [58], whereas classical results rely on the work of Silverstein [59]. Moreover, in addition to the linear filters studied in [55], we study the enhancements provided by the optimum and successive interference cancellation filters.

5.2 Model

We consider a single isolated cluster. The spreading length is denoted N . The number of T-nodes in the cell is K . The load is $\alpha = K/N$. The model is essentially the same as in the previous chapter, removing the inter-cluster interference term. For this model, asymptotic expressions of the SINR β_k of user k can be derived.

5.3 Games and Equilibria

A game with a unique strategy set for all T-nodes is defined by a triple $\{S, \mathbb{P}, (u_k)_{k \in S}\}$ where S is the set of *players*, $\mathbb{P}$ is the set of *strategies*, and $(u_k)_{k \in S}$ is the set of *utility functions*, $u_k : \mathbb{P}^{|S|} \rightarrow \mathbb{R}$.

In our setting, the players are simply the T-nodes, indexed by the set $S^K = \{1, \dots, K\}$. The strategy for a T-node is its power allocation P_k , which we will assume belongs to a compact interval $\mathbb{P} = [0, P_{\max}] \subseteq \mathbb{R}$. The utility measures the gain of a T-node as a result of the strategy this T-node plays. In [56], the author derives what he calls Throughput to Power Ratio (TPR) under minimal requirements. The utility of T-node k is expressed

$$u_k = \frac{\gamma_k}{P_k}. \quad (12)$$

We denote $\gamma_k = \gamma(\beta_k)$, where $\gamma(\cdot)$ is the same function for all T-nodes. In (12), γ is at least C^2 and should satisfy conditions detailed in [56] in order to obtain an “interesting” equilibrium.

This utility is expressed in *bits per joule*. In the non-cooperative game setting, each T-node wants to maximize its utility to the detriment of the others. A Nash equilibrium is obtained when no T-node can benefit by unilaterally deviating from its strategy.

To obtain the maximum utility achievable by T-node k , we show that β_k has to be the solution of the equation

$$\beta_k \gamma'(\beta_k) - \gamma(\beta_k) = 0. \quad (13)$$

The existence of a solution to (13) is guaranteed as long as the function $\gamma(\cdot)$ is a quasiconcave function of the SINR, i.e., there exists a point below which the function is non-decreasing, and above which the function is non-increasing [60, 56].

5.4 Power Allocation in the Nash Equilibrium

The results of [55] for Matched and MMSE filters are a special case of our setting when $L = 1$ (flat fading case). In addition, we derive the power allocation for the Optimum filter.

We show that the influence of the strategy of a player on the payoffs of other players is (asymptotically) “small”. It justifies the fact that we can obtain an equilibrium in the asymptotic setting, without the need for players to possess all the information on the system. Their local information is sufficient. In the asymptotic limit, we obtain results similar to the Wardrop equilibrium: the strategy used by each T-node does not influence the strategy of other T-nodes.

In the context of frequency-selective fading, for each T-node k , denoted by its index $\frac{k}{N} = x \in [0, \alpha]$, there are $L > 1$ paths with respective attenuations $h_\ell(x)$, $\ell = 1, \dots, L$, which are i.i.d. random variables with some known distribution. Given the target SINR β^* , the Nash equilibrium power allocation is determined by implicit equations for the various receivers.

5.5 Successive Interference Cancellation

The optimal filter provides a bound to the performance that can be achieved through (non-linear) filtering at the U-node. In order to improve the performance of the system, we introduce Successive Interference Cancellation (SIC) [61] at the base station. Under the assumption of perfect decoding, SIC immensely improves the performance of linear filters (Matched Filter or MMSE Filter). The MMSE SIC filter actually achieves the optimum filter bound, under the assumption of perfect decoding. The principle of SIC receivers is quite simple: T-nodes are ordered and are decoded successively. At each step, supposing that the T-node has been encoded at the appropriate decoding rate, the signal is decoded and its contribution to the interference is then perfectly subtracted. This removes some of the inter-T-node interference and therefore increases the SINR of the following decoded T-nodes.

The challenge is that the T-nodes must transmit at the appropriate rate to avoid the catastrophic occurrence of imperfect decoding. Usually, the ordering of T-nodes is done in a centralized way, at the base station which then communicates it to the T-nodes. However, for the protocol to remain distributed, T-nodes should be able to decide, based on their local information, at which rate to transmit. At equilibrium, the rate is determined by the SINR β^* , and it is the transmission power of the T-node that is determined according to its rank of decoding. In the following, we assume that each T-node has a unique i.d. number j ranging between 1 to K . Using the correlated games framework, the U-node serves as an arbitrator that allows a distributed ordering of T-nodes. This gives rise to a different kind of power allocation, which depends explicitly on the order in which the T-nodes are decoded.

Once the T-nodes know their order, they can calculate their transmit power according to the filter that is used. The equilibrium still occurs when all T-nodes reach the SINR β^* . A single T-node will not benefit by deviating from this, since it would decrease its utility.

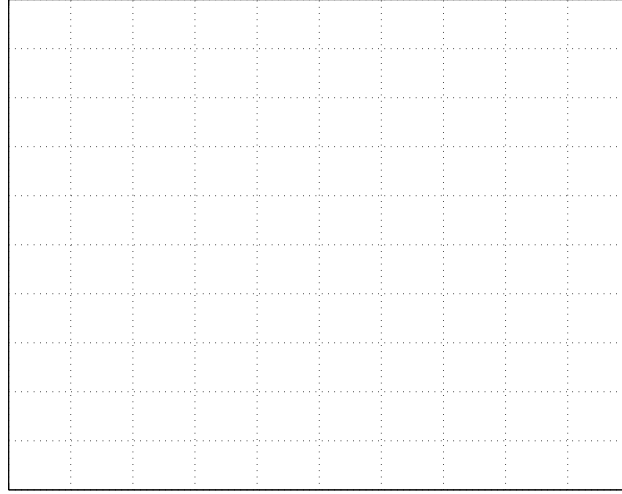


Figure 10: Average inverse power used by different filters

5.6 Numerical Results

In order to visualize the results obtained with the different filters, simulations are plotted for a CDMA system with K users and a spreading factor $N = 256$. The load of the system is $\frac{K}{N} = \alpha$. The noise variance is $\sigma^2 = 10^{-10}$. For a number of bits in a CDMA packet $M = 100$, the goodput is $\gamma(\beta) = (1 - e^{-\beta})^{100}$, and $\beta^* = 6.48$. The capacity achieved at the Nash Equilibrium is $C = \alpha \log_2(1 + \beta^*) = 0.39$ bits/s.

In Fig. 10 we have plotted the average of the inverse power of the users in the Nash equilibrium for each of the investigated filters in function of the load α . We plot the average inverse power because of the direct relation to the utility for the users. The higher this average, the higher the utility for the user. The SIC filters are always more efficient than their linear counterparts. In addition to the theoretical curves, Monte-Carlo simulations were performed both with random ordering (circles) and ordering by decreasing total channel energy (crosses), for $L = 8$ multipaths. Simulations show that the optimal ordering improves the power efficiency of the successive interference cancellation filters.

5.7 Conclusion

Using the tools of random matrices, we have derived the equilibrium power allocation in a game-theoretic framework applied to an asymptotic multi-user system with cyclic prefix, under frequency-selective fading. Three receivers are considered: matched filter, MMSE and optimum filter (given by Shannon's capacity). In addition, distributed ordering mechanisms are introduced and the successive interference cancellation variants of the linear filters are studied. For each T-node, this power allocation depends only on the total energy of the channel of the T-node under consideration. For a frequency-flat channel, the power allocation among T-nodes is dis-uniform, whereas when the number of multipaths increases, the power allocation increasingly becomes a

uniform one.

6 The BIONETS simulator platform

6.1 Benefits

The vision of the simulator is to provide a simulation framework that can be used by every BIONETS partner. It already includes some of the algorithms, concepts from the entire BIONETS community. Every design approach, be it Formal Modeling or Agile Programming, tries to catch errors early during the design process to reduce the probability of late discovery of design flaws. This is why it is so important to get early simulation results for the BIONETS network.

6.2 Goals

The main goal of the simulator is to provide a tool for all participants, so they can access the implementations of early ideas and algorithms from any Work Package. Another goal is to help experiment with novel ideas that may be implemented in BIONETS. Such topics are:

- Information dissemination algorithms
- Information filtering (Semantic Processing, Redundancy Management)
- Active networking
- Changing user needs
- Evolving services (Genetic Algorithms, Genetic Programming, Chemical Programming)
- Social/Economic Concepts (Information Bartering, Trust and Credibility Management)
- Information dependent mobility models.

Each of these technologies and approaches should be neatly and seamlessly integrated into a common simulation framework. It is also crucial to isolate the functionality of these modules, to help independent development. These challenges of the Simulator are not always necessarily the challenges of the BIONETS project, and vice versa. For BIONETS the challenge is to answer the *'How to implement?'* question, while for the simulator is *'How to help the community to try out many ideas?'*.

6.3 Realization

In this subsection the fundamentals that were followed during the development of the BIONETS framework for OMNeT++ are described. The architecture of the simulator has to be general and flexible enough to meet the diverse requirements. It also important to support quick prototyping, so participants can easily and quickly add their own ideas to the simulator. For this, the simulation framework provides modules with standardized interfaces that hide most of the implementation details. The simulator code is organized into modules that can be easily assembled into complex models. This approach promotes code reusability, helps testing and isolated development.

Because of the fact that the collaboration of the whole community has great benefits, the simulator has a great emphasis on quality documentation and developer communication. To achieve this, many tools have been made available to help efficient teamwork: website, wiki site, mailing list, and a version management system integrated with a bug-tracking system.

The documentation includes the following: User's Manual, Quickstart Guide, Tutorials and API Documentation. The tutorials and the quickstart guide help the user get started with the usage of the simulator. Those describe an example project step-by-step that the user can follow and try. The User's Manual is a complete description of the simulator and its tools. It has a linear structure and users are intended to read it completely. The manual is not as detailed as a reference because it is difficult to remain up-to-date with minor changes, but it is more readable, as it is designed primarily for learning. While the User's Manual is for learning, the API Documentation is a reference for daily use.

There is an installer package (downloadable from the website) that is currently available only for Windows users, but the simulation framework itself can be used in Linux also. In this case one must install the libraries and other dependencies manually. The simulator package contains the tools needed on Windows: OMNeT++, SCons, and Python for the building, and TortoiseSVN for accessing the repository. On Linux systems the only difference is that the SVN client is used instead of TortoiseSVN.

6.4 The simulation framework

6.5 Main structure

The simulator is designed to support quick prototyping and the swift realization of ideas. There is also a great emphasis on flexibility. The simulation framework itself is a collection of interdependent software components. The heart of the system is the OMNeT++ [62], [63] discrete event simulator library and tools. The Mobility Framework [64], [65] is a library built on the OMNeT++ core library which provides a framework for mobile communication simulations. It contains several implemented modules. Simulator components provide additional or customized services on top of OMNeT++ and Mobility Framework.

OMNeT++ itself has several components. The core library is the implementation of the discrete event simulator which executes a simulation model that can be defined by the simulator users.

The mobility framework is a library that provides a framework for mobile ad hoc simulations and also has different model implementations. It supports the use of different mobility and channel models, and many network technologies.

The main component of the simulator is the Module Library which places the BIONETS model framework on top of the Mobility Framework. It also contains the implementation of BIONETS algorithms and theories. The architecture of the Module Library must be designed very carefully as BIONETS is under active research and changes in concepts are possible. The simulator is fully functional only with the Module Library. Further modules such as visualization components,

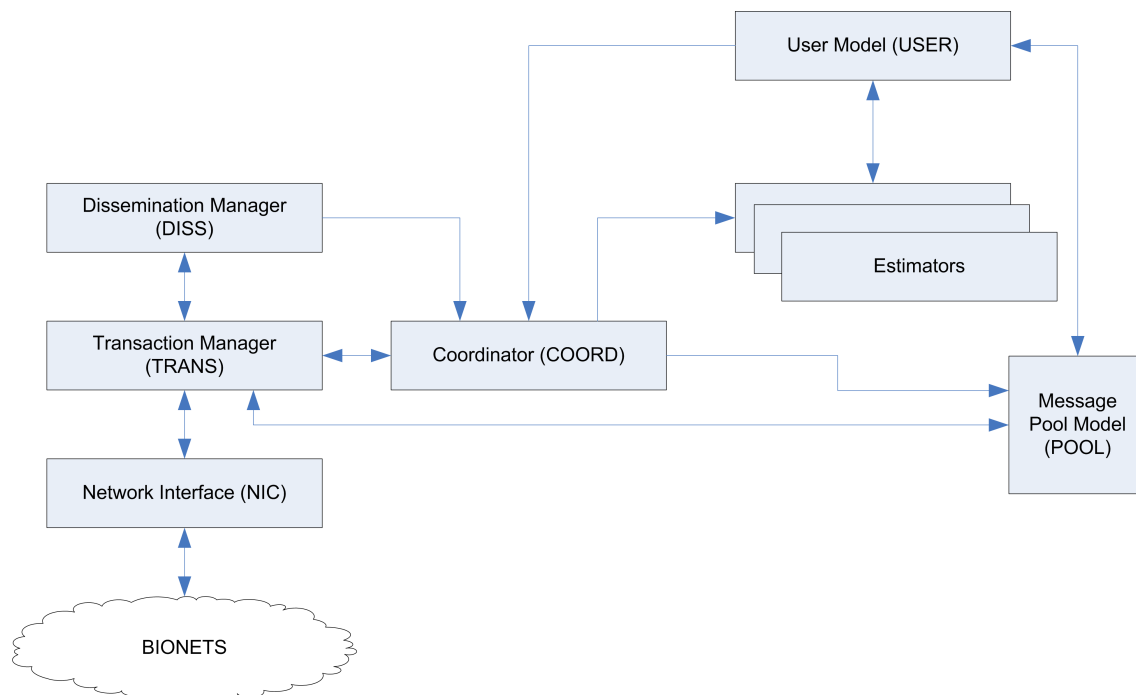


Figure 11: Architecture of the U-Node implementation

scenario editor and numerical tool integration provide services that make the simulator easier and faster to work with.

6.6 Module Library architecture

Because the simulator is deployed quite early in the BIONETS project, the library design is crucial. The simulator Module Library must face emerging, non-traditional ideas during the entire lifespan of the project. On the other hand, one goal of the simulator is to assist the early prototyping and quick evaluation of ideas. These two goals together requires a very flexible design with high code reusability.

In reality even with a very careful design, the rewriting and refactoring of software components is unavoidable. The design of the Module Library is intended to minimize and isolate these refactoring tasks.

The architecture, ideas and structure of modules in this document do not necessarily represent the structure of a real-world implementation. The Module Library regards the modeling and simulating the system, instead of exactly implementing it. It also must be as general as possible, supporting even the most complex and challenging ideas, whereas an actual implementation is optimized for the final BIONETS system.

One problem with a general architecture is that it makes rapid prototyping harder. One needs to implement numerous unneeded functionalities just to test one specific function. To avoid this the Module Library uses Null Modules which are modules that implement a trivial or stub functionality, but does not violate the general architecture goals.

Currently our efforts are limited to the modeling of the U-Nodes as the most complex entities of the BIONETS system. Figure 11 shows an overview of the architecture. The U-Node communicates with other nodes through the Network interface (NIC). The Transaction Manager module (TRANS) handles the message exchange between nodes, which are called transactions. The Dissemination Manager (DISS) is responsible for locating available nodes that are in need of some service. The task of the Coordinator (COORD) module is to initiate transactions (exchange of messages) when the conditions are desirable, and to regulate the operation of DISS and TRANS modules. When a message arrives, the COORD module consults with the Estimator modules on how useful the information seems to be. The Internal Value Estimator (IVEST) estimates the importance of the message for the user. The External Value Estimator (EVEST) module estimates how useful the information held by the message for other user is. This is necessary when messages are bartered between U-Nodes. REDEST or Redundancy Estimator deals with estimating the redundancy of a new message relative to already-received messages. Message Pool (POOL) is a database of stored messages.

We have used the simulator to evaluate the performance of different information dissemination algorithms and mobility models. To achieve this, we have elaborated the DISS and TRANS modules and some mobility algorithms discussed in the sections which follow.. For the other modules we used some basic implementations that may be improved upon later.

All of the WP 1.3.2 partners have become familiarized with the simulation framework and denoted that they would like to contribute to the Module Library and apply the simulator to their scenarios. We have also encouraged all members of other Work Packages to get acquainted with the simulator, by using it to create their modules and scenarios, and get results from them. For example in the future Fraglets, a novel programming language introduced in D2.2.2, will be implemented in the simulator.

6.7 Reference algorithms for information dissemination

Our decisions on the selected algorithms are primarily based on the survey of Williams et al. [66] and the results from Dai et al. [67]. The first one covers a large set of different dissemination types and gives a very good overall picture of the existing solutions. [67] on the other hand concentrates mainly on the performance of the different self-pruning algorithms.

Several dissemination methods were investigated, including the probability and counter-based [68] methods, Generic Self-Pruning [69], Scalable Broadcast Algorithm (SBA) [70], Multipoint Relaying (MPR) [71], Ad Hoc Broadcast Protocol (AHBP), Lightweight and Efficient Network-Wide Broadcast (LENWB), HyperGossiping [72], Wu and Li's algorithm [73], Stojmenovic's method [67, 74], Optimized Flooding Protocol (OFF) [75] and Distance Adaptive Dissemination (DAD) [76].

All of the location-based algorithms were excluded for numerous reasons. One of them is that we cannot expect that a large number of U-Nodes will be equipped with GPS receivers. The other problem is that the papers that propose these algorithms do not take into account the precision of the currently available positioning systems. While this may or may not be a problem, we

followed a conservative path, and excluded these algorithms entirely. The algorithms that rely on signal strength as a measure of distance were also left out, as we contend these this can be very unreliable in an urban environment.

Several methods were too asymmetric, in the sense that they rely on the very specific behavior of some of the nodes. These algorithms do not fit well in the BIONETS approach, given the fact, that in the future dissemination methods may be acquired through the use of genetic programming (GP). In these scenarios it is quite possible that nodes refuse to be cluster heads, message cache servers or to perform a specific task that gives no gain to the node itself. This of course excludes the neighbor designation methods, like AHBP [77].

Williams et al. [66] compared several dissemination methods in their work which included simple heuristic-based methods, a location-based scheme and two self-pruneing methods. They simulated these protocols under different network conditions such as varying traffic load and mobility. From the results of their experiments we came to the conclusion that the best available dissemination methods are self-pruning based, and the Scalable Broadcast Algorithm (SBA) is a very good all-around solution that performs well in a wide range of situations.

In our further research we tried to focus on self-pruning methods. As it turns out from [67], most of these algorithms cope well with the problem of congestion; in almost all cases a 98% delivery ratio was reached. Mobility is a much greater threat to these algorithms. Most of the self-pruneing algorithms perform poorly if the nodes in the network move at high speed. It has also been shown that the algorithms that rely on node neighbor degree instead of node ID are less resilient to mobility, and their performance degrades more quickly than the other methods'. From the comparison three algorithms were chosen that showed good properties. SBA is outperformed by some of the other methods, but it showed remarkable resiliency to extreme mobility conditions. The Generic self-pruneing method, introduced by the authors in an earlier article, was the most effective algorithm in many situations, but somewhat sensitive to mobility. However this sensitivity can be eliminated if the algorithm uses location information like the Stojmenovic method.

HyperGossip is a relatively new information dissemination method, and there are no known performance comparisons with other algorithms. It was a straightforward selection because BIONETS networks will be also partitioned, so this approach fits in very well.

6.8 Scalable Broadcast Algorithm (SBA)

The SBA algorithm (introduced in [70]) is a neighbor-based, self-pruning protocol. It requires 2-hop neighbor information and the last sender ID in the broadcasted packet. To acquire the neighborhood information, it sends periodically HELLO packets which contain a list of the sender's 1-hop neighbors.

SBA has an explicit Random Assessment Delay (RAD) mechanism. This means that the packets are not forwarded immediately, but after a random backoff interval. During this interval the nodes monitor their surroundings. When a node v receives a broadcast packet from a node u it excludes the neighbors of u , $N(u)$ from the set of his own neighbors $N(v)$. The resulting set

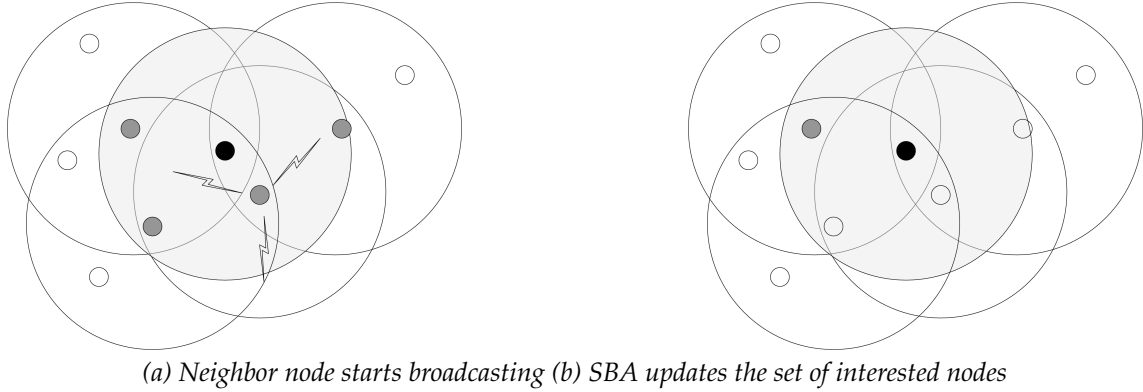


Figure 12: Demonstration of SBA context update when neighbor node broadcasts

$B = N(v) - N(u)$ is the set of the potentially interested nodes. This mechanism is demonstrated in figure 12. If all of its neighbors are covered before the RAD interval is over, the node cancels the rebroadcast.

While the length of the RAD interval is random, the interval from which it is chosen is not. For a node v the maximum RAD is calculated by the $\left(\frac{d_v}{d_{max}}\right) \cdot T_{max}$ formula, where $d_v = |N(v)|$ and d_{max} is the degree of the node with the largest degree in $N(v)$. Nodes choose the time of transmission uniformly from this interval. This ensures that nodes with higher degree usually broadcast before nodes with fewer neighbors.

6.9 Generic Self-Pruning

Generic self-pruning was introduced first as two general self-pruning rules in [69] and then a specific version of the rules as used in [67] for comparison with other algorithms. In its general form it relies on k-hop neighborhood and k-hop routing information. The class of algorithms they describe use one of the versions of the so-called Coverage Condition.

When a node receives a message, it decides whether to forward it or not, based on the Coverage Condition:

Coverage Condition I (static) Node v has a non-forwarding status if for any two neighbors u and w a replacement path exists that connects u and w via several immediate nodes (if any) with either higher priority values than the priority of v or with visited node status.

Generic self-pruning contains many existing algorithms as special cases of Coverage Condition I or II (both of them is detailed in [69]).

6.10 Hypergossiping

Hypergossiping [72] is an algorithm specifically designed for partitioned networks, where nodes are mobile, and partitions join and split from time to time.

The algorithm uses a simple adaptive gossiping strategy for in-partition forwarding, but re-broadcasts some of the packets if it detects a join with another partition. The join detection is

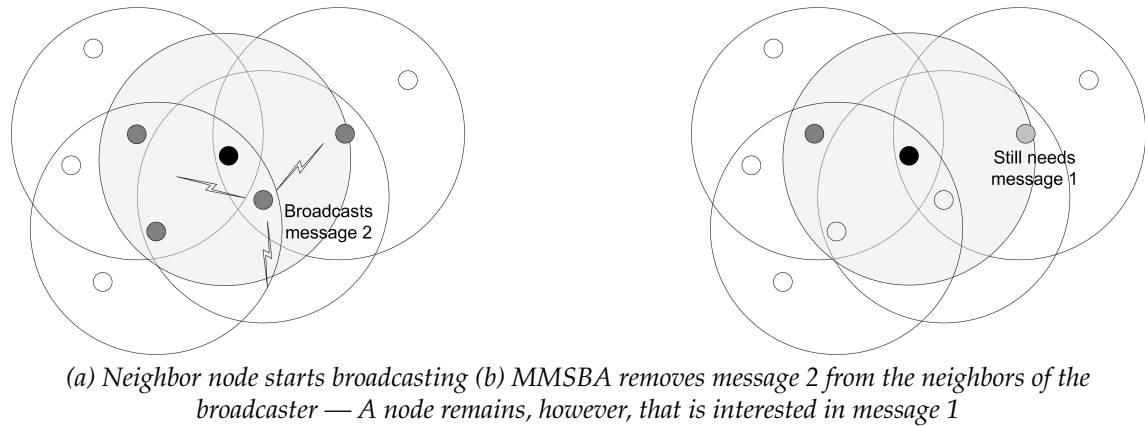


Figure 13: Demonstration of MMSBA context update when neighbor node broadcasts

based on a simple heuristic, that nodes in the same partition have recently received the same messages.

Every node maintains a list, called LBR (Last Broadcasts Received), of the recently broadcasted messages. They send HELLO messages periodically, to indicate their presence. When a new node is detected, one of the nodes includes its LBR in the next HELLO message. When the other node receives this LBR, it compares it with its own LBR. If the overlap between the LBR of two nodes is smaller than a threshold, a new message is sent, called BR (Broadcasts received), which contains the list of messages that the nodes have already received. From this the other node knows, that a partition join has happened, and rebroadcasts all the messages that have not yet been received by the other node.

After this rebroadcast, further dissemination happens using adaptive gossiping.

6.11 New algorithms developed in WP1.3

6.12 Multi-Message SBA (MMSBA) — An improved version of SBA

The algorithms that do not take partitioning into account, like SBA and Generic, perform poorly in disconnected environments. These algorithms rebroadcast a message only once, when it is first received. This means, that if there are no more nodes in transmission range when the last message is sent, the dissemination stops. When a new node arrives into the transmission range, there is no event that can trigger a retransmission. To solve this problem, we have chosen the SBA algorithm for improvement, because it is computationally simple, and proved to be robust according to prior research [67, 66]. It also gives a nice way to generalize its mechanism to the multi-message case.

The basic idea of the SBA algorithm is the RAD process, which delays the transmission of packets by a random interval. The first time SBA receives a packet, it starts the RAD procedure. During this interval the SBA listens to its neighbors. When one of them starts broadcasting, SBA removes the neighbors of the broadcasting nodes from its own 1-hop neighbor set. Our modified algorithm, the Multi-Message SBA (MMSBA) triggers a RAD, not only on the first reception of a

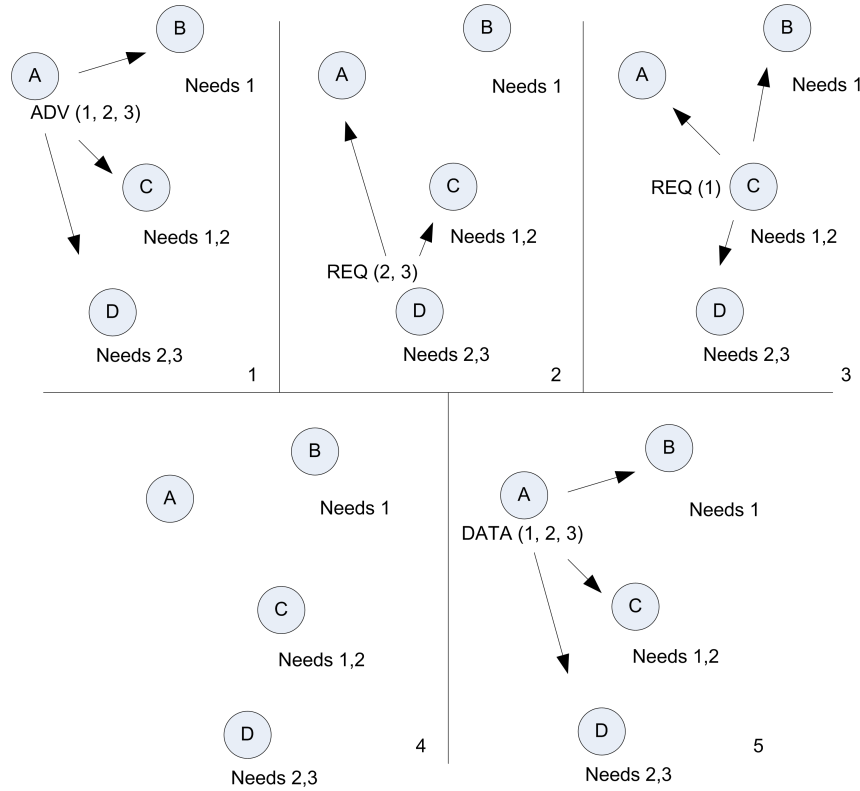


Figure 14: IOBIO handshake sequence

message, but on any event that changes the local neighborhood information. Every time a HELLO message is received, MMSBA updates its neighbor list. When the number of interested nodes is larger than zero, MMSBA starts the RAD. The RAD works exactly the same way as in SBA.

A minor problem can be that every time we receive a HELLO message from another node, we add it to the list of interested nodes, even if the broadcast message was already sent to them. This problem is not present in the original SBA, as nodes broadcast the message at most once, when it is first received. To overcome this problem, the nodes include the list of messages they have already received in their HELLO packets. This also provides a feedback to MMSBA if a broadcast message was lost during transmission.

To support multiple messages, the RAD process also needs to be updated. The new method is demonstrated in figure 13. One of the neighbors starts broadcasting message 2 during the RAD. Here MMSBA removes from its context the nodes that are interested *only* in message 2. However, nodes that are interested in other messages (for example message 1) remain in its list. This mechanism allows MMSBA to handle many messages independently.

6.13 A new algorithm: Modified IOBIO (InfOrmation Dissemination Protocol for BIOlogically Inspired autonomic Networks and Services)

The IOBIO algorithm was introduced in D1.3.1, and it is specially designed for the BIONETS architecture. It uses a simple 3-stage handshake to discover neighbors that are interested in the

Parameter	Value
Area	500 × 500
Carrier frequency	2.5 GHz (ISM)
Thermal noise	-110 dBm
Pathloss coefficient	3.2
Bitrate	1 Mbit/sec
Duration	10 sec (with 200 sec warm up)
Number of runs	100 / point

Table 1: Common simulation parameters

message. The goal of the protocol was to reduce an unnecessary load of nodes by duplicate or unneeded data ("spamming").

There are three IOBIO message types that are used by the protocol. The ADV (Advertisement) messages are sent periodically and they contain the list of messages in the possession of the sending node. In our modified version the period of the advertisement is adjusted adaptively, based on a simple heuristic. When another node broadcasts ADV packets around the node, the period of the advertisement is increased. This works very well in a partitioned environment, where the chance of reaching an additional node decreases over time.

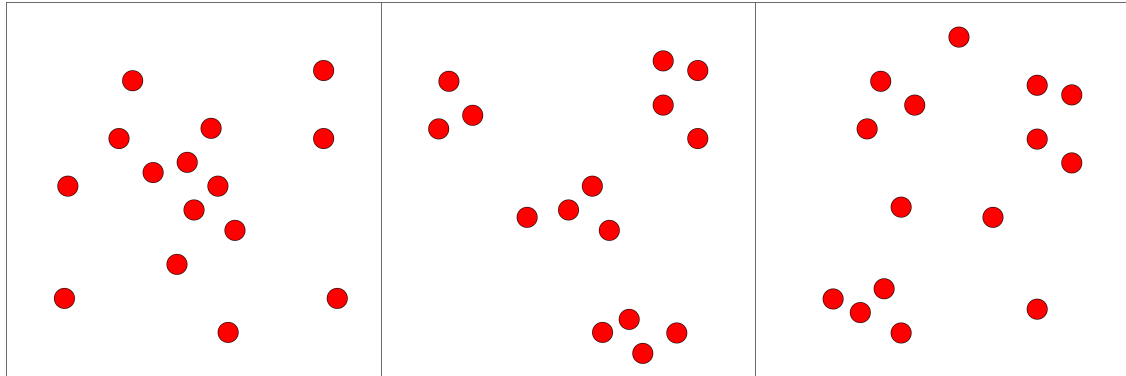
Neighbor nodes indicate their request by sending an REQ (Request) packet and the originator node sends the DATA packet to them. The transmission of a REQ after an ADV is not done immediately, but after a random delay. This process is demonstrated in figure 14. Node A sends an advertisement indicating that it has the messages 1, 2 and 3. After receiving this ADV, nodes B, C, D start a random delay. At step 2 node D sends a REQ packet, indicating, that it needs messages 2 and 3. At step 3, node C sends a REQ packet. However, sensing the REQ packet of D, it knows that message 2 is already requested, and only puts the ID of message 1 in the REQ packet. At step 4, the random delay of B is over; however, message 1 is already requested, so no REQ is sent. At step 5 the wait interval of node A is over, and it broadcasts all requested messages.

6.14 Simulation setup

The main criteria for the evaluation of the algorithms was the robustness of the algorithms in different conditions. The simulation parameters were chosen to make it realistic to be able to test the algorithms in difficult environments.

The common parameters of the simulations are shown on Table 1. It is assumed that there is no Forward Error Correction (FEC) implemented in the nodes, but nodes are able to detect errors in the packets and drop them. Data packets are assumed to be 4096 bytes long, while control packets have changing length depending on the information they carry (like neighbor information). Node and message identifiers are 64 bits long, and there is a 16 byte header in the control messages.

Every simulation starts with a 200 second "warm-up" period, which is required by the mobility models. The transmission of the control messages (like HELLO) starts after the 195th second, so the dissemination methods have time to discover their neighbors. The first message is broadcast from a single node at the 200th second. It is assumed that the message has a 10 second



(a) Constant Speed Mobility (b) Reference Point Group Mobility Model (c) Reference Point Group Mobility Model with Dynamic Groups

Figure 15: Examples of different mobility models

lifetime, so the simulation stops at 210 seconds, when the message gets outdated. Every point in the graphs was averaged over 100 runs.

6.15 Mobility models

We used three different mobility models that have different properties. All of these models are described in D1.3.1.

6.15.1 Constant Speed Mobility

A variant of the commonly used Random Waypoint Mobility is [78]. Nodes uniformly choose a random point on the map and move towards it, until they eventually reach it. Then the node chooses another waypoint, goes toward it until it reaches it, and so on. The difference between our model and the original Random Waypoint Mobility is, that the nodes in our model does not pause for random intervals, and the speed of the nodes is constant. This is not a big difference, as using the original random waypoint mobility most of the nodes travel at the same speed anyway (close to the slowest speed possible in the model).

It is known, that using this mobility model, the nodes tend to be in the central areas of the network. An illustration of this can be seen in figure 15.

6.15.2 Reference Point Group Mobility Model (RPGMM)

In this model the nodes are organized into groups and the members of the groups move together. The nodes choose their group at the start of the simulation, and then remain in that group for the entire duration. Each group has a center point that moves according to a mobility model (in our case the Constant Speed Mobility Model). The nodes choose their destination uniformly around the reference point. This model makes the simulated network inhomogeneous and highly partitioned. An illustration can be seen in figure 15.

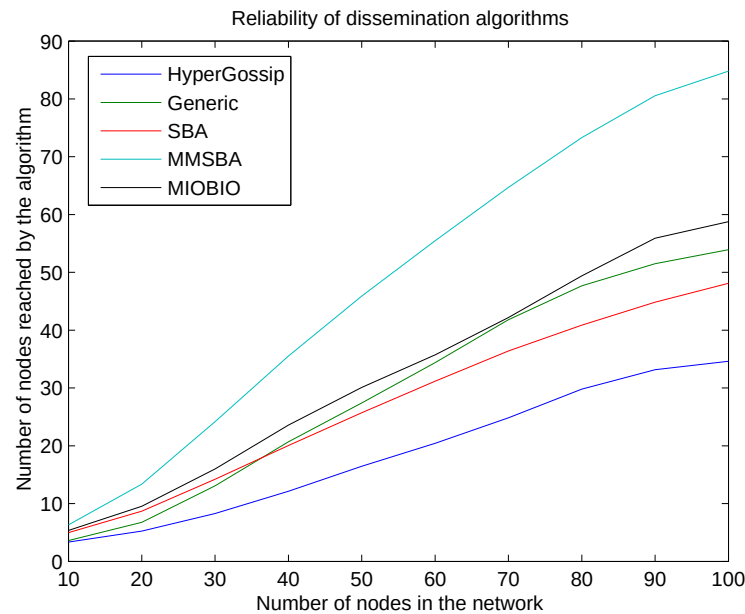


Figure 16: Reliability of the simulated algorithms using Constant Speed Mobility

6.15.3 Reference Point Group Mobility Model with Dynamic Groups (RPGMMDG)

This is a modified version of the RPGMM. Instead of choosing a group at the start of the simulation, the nodes change their groups from time to time. This makes the network more connected, as there are always nodes wandering in the space between the high density regions, thus helping transfer of data between groups. This makes this model much like a hybrid between the two previous models.

6.16 Simulation results

For each algorithm we measured the number of the nodes that were covered during the lifetime of the message (10 sec). The results of these measurements are shown in figures 16, 17, 18. It is clear that the mobility patterns affect the performance of the dissemination algorithms; all of them turns out to be less reliable when group mobility is used. The performance drop is the largest when the RPGMM model is used. In this case the network is prone to partitioning, and in islands, where node density is very high, congestion can lead to message loss. The results using the RPGMMDG model are a bit better, because there are always nodes wandering between the different groups. This helps to connect the dense islands.

The results show that MMSBA provides the best coverage for the Constant Speed Mobility, but using other models, the performance of MIOBIO and MMSBA is very close. Both of the new algorithms outperform SBA, Generic and HyperGossip.

An important property of the algorithms is the overhead needed to send one byte of data to other nodes. We measured the overhead from two perspectives. We measured the ratio of total sent bytes (including control packets) and received useful data (non-duplicate data) and the ratio

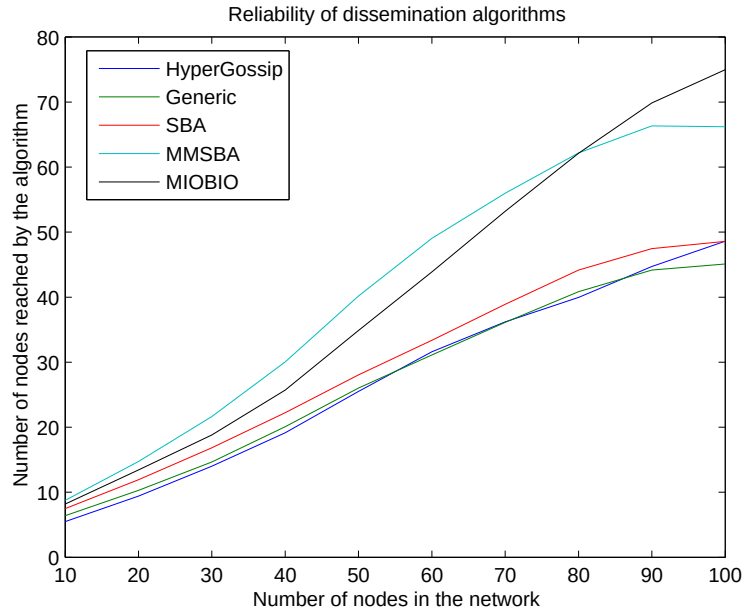


Figure 17: Reliability of the simulated algorithms using RPGMMDG

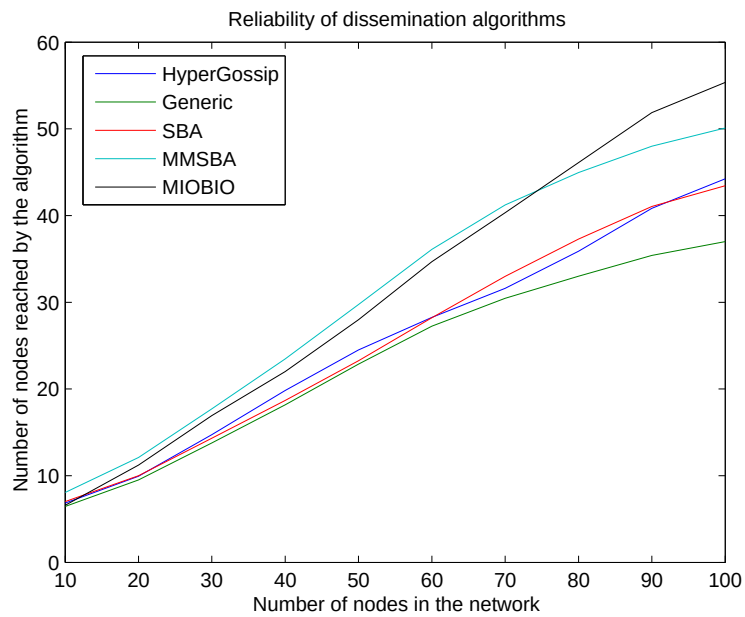


Figure 18: Reliability of the simulated algorithms using RPGMM

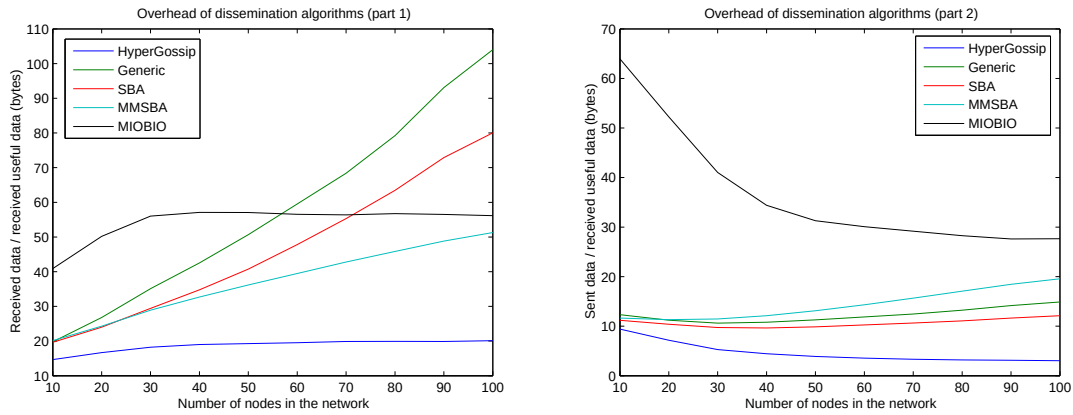


Figure 19: Overhead of dissemination algorithms using Constant Speed Mobility

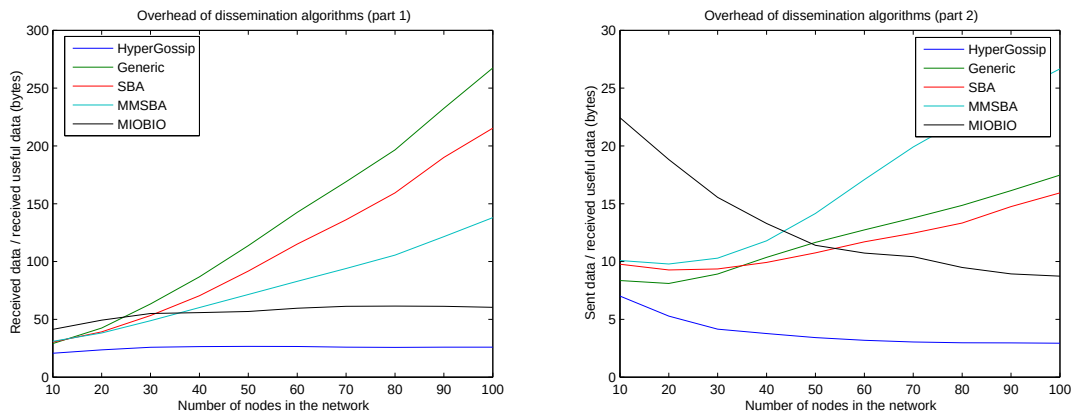


Figure 20: Overhead of dissemination algorithms using RPGMMDG

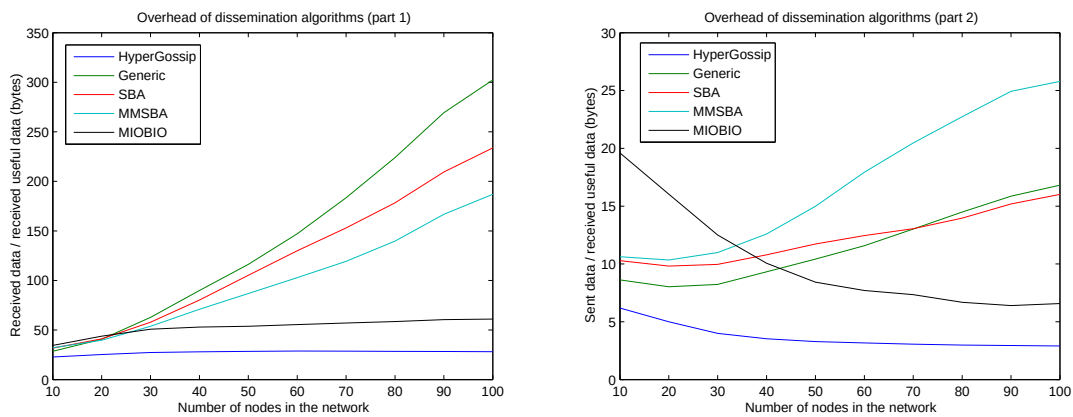


Figure 21: Overhead of dissemination algorithms using RPGMM

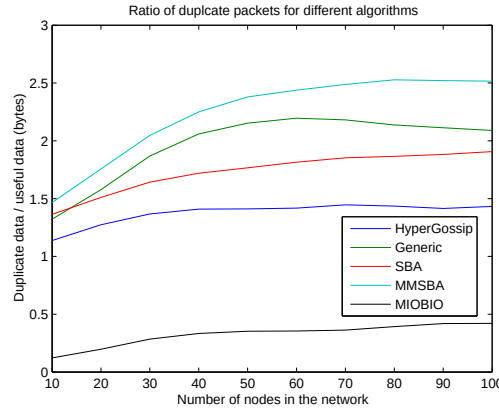


Figure 22: Ratio of duplicate packets using Constant Speed Mobility

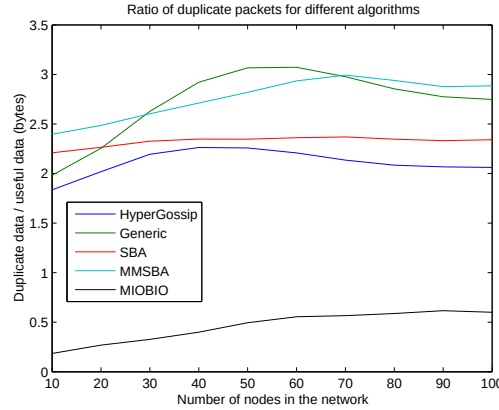


Figure 23: Ratio of duplicate packets using RPGMMDG

of received bytes and useful data. The first indicates the battery usage of the sender nodes, the latter indicates channel utilization efficiency. The results are shown on figure 19-21.

It is clear from the results, that the mobility models influence the performance of the algorithms. HyperGossip has the lowest overhead in all of the cases, but it also has the lowest node coverage. MMSBA performs best when Constant Speed Mobility is used, but it gets more inefficient when the network is more partitioned. MIOBIO on the other hand has very low overhead in the partitioned cases. We can also observe, that its overhead values scale much better than the values of MMSBA.

We are also interested in the number of duplicate data packets too, because these must be processed by the Information Filtering module in U-Nodes, and this can be a serious computational overhead. The results of the simulations are shown in figure 22-24. In this case MIOBIO is the clear winner, as it is able to maintain the 1:2 duplicate/useful data ratio independently from mobility models.

To test the performance of the algorithms when multiple message types are present, we evaluated a multi-message scenario. In this scenario, every node has an initial set of messages at the

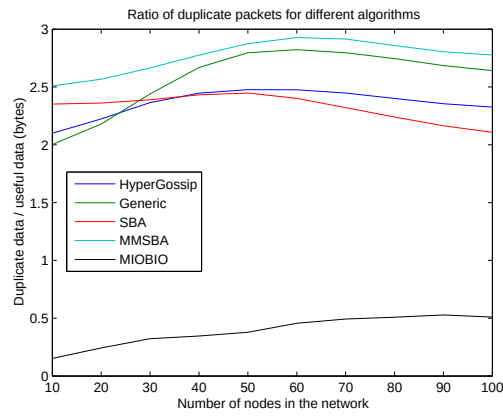


Figure 24: Ratio of duplicate packets using RPGMM

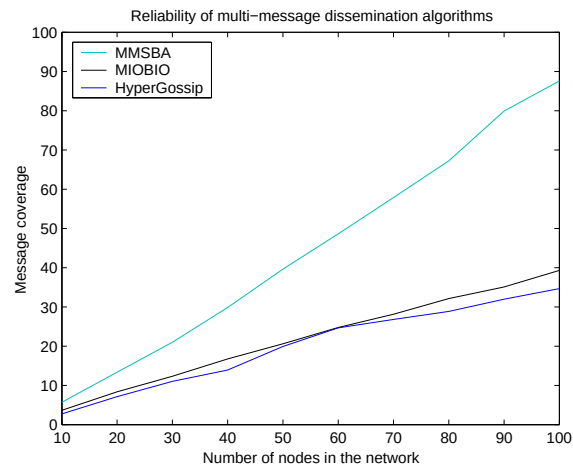


Figure 25: Reliability of dissemination algorithms using the multi-message scenario

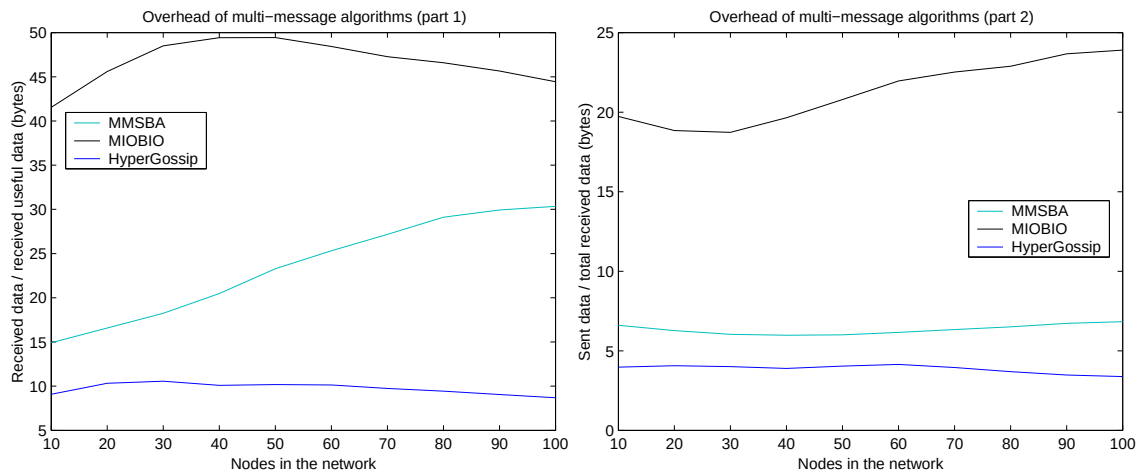


Figure 26: Overhead of dissemination algorithms using the multi-message scenario

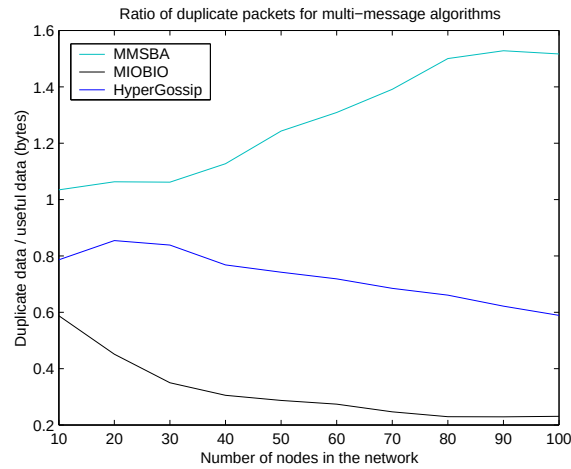


Figure 27: Ratio of duplicate packets using the multi-message scenario

beginning of the simulation. The size of this set is chosen randomly from 0–2 for each node, and the type of the initial messages are chosen randomly from 5 different types. After the simulation starts, every node tries to disseminate the messages it has, while it also tries to get the missing ones.

With these parameters only HyperGossip, MMSBA and IOBIO were simulated, as these are the only ones that support the dissemination of multiple messages. In figure 25 the reliability of the three algorithms is shown. The diagram shows the total number of messages that the nodes have at the end of the simulation divided by the amount of message types, so the results can be compared to the earlier figures. It is clear that MMSBA is particularly good at disseminating multiple messages from multiple sources, and is almost twice as reliable as the others.

In figure 26,27 the different overhead measures are shown. The HyperGossip shows again the lowest overhead, despite its very poor delivery rate. The overhead of IOBIO is the worst, only the rate of duplicate data messages is better than the others.

7 Simulations of situation-aware opportunistic routing

7.1 Introduction and motivation

The starting point for the VTT *self-organization* research line within BIONETS SP1 is based on the assumption that the dynamic life of each BIONETS node (U/T) and their composition and mobility require intelligent situated and adaptive functionality. The requirements for such a self-organization research line has been studied and documented within WP1.1 deliverables. According to the requirements each node shall be aware of itself and its environmental situation. When some change occurs the node should react in accordance with the change situation. This self-organization is related to changes, but the changes should not be like ad hoc route changes in the connected ad hoc networks. This is because in the BIONETS type of disconnected opportunistic system, the source and destination nodes are not necessarily available in the same network simultaneously. However, the mobile nodes are required to communicate even if no route can be discovered and the topology of the complete network is not known by each node. Thus other types of routing schemes are required than connected ad hoc networks.

One potential starting point has been the model of Delay-Tolerant Networks (DTN) [?], [?] which introduces a disconnected type of network where the expectations of node resources are limited and the messages are forwarded asynchronously. The architecture also includes the concept of DTN gateways between isolated internet clusters. However, in BIONETS case each node shall determine what to do with the incoming message and in fact each BIONETS node (most likely an U-node) shall act as a gateway. When the message comes in, the node decides the next hop according to the specific context, to which the node is adapted (situationally adaptive). The opportunistic routing can be categorized as routing with or without infrastructure and multi-tier routing. Some examples of these algorithms are described in 6.7. The challenge of situated adaptive behavior requires the application of self-organized context-based type of opportunistic routing schemes.

The BIONETS system may have a connection point towards fixed infrastructure, and the referred connection point can be mobile which refers to the challenge of mobile infrastructure. In the mobile infrastructure case some specialized nodes can act as carriers, ferries or mules of data towards the network and servers where the data is destined. This refers to the challenge of inter-operation with possible fixed traditional infrastructure networks. In addition, the BIONETS system may contain multiple clouds of different kinds of self-organized network clusters, and therefore it can be termed a self-organized multitier opportunistic network which may or may not work together with fixed networks.

In summary, this section will focus upon the simulations of situated opportunistic routing which enables situated adaptive self-organization for messaging between specific BIONETS nodes, the requirements of which are described within WP1.1 and specifications described within WP1.2. The architectures of the U and T-nodes as well as the optional Access Points are described in detail in other deliverables. In addition the algorithms and protocol definitions are excluded from the scope of this document. Later in this chapter discussion of the employment of situated opportunistic routing in OMNet++ is included in section 7.2. The scenarios and mobility models

utilized are presented in section 7.3. Descriptions of the simulations done thus far are provided in section 7.4 and results obtained from the simulations are presented in section 7.5.

7.2 Model structure

The implementation of a node in the simulator has been built on top the modular design of the OMNeT++ and Mobility Framework. It allows the system to be constructed from small modules by incorporating N number of simple modules into one composite module. These composite modules will carry out the duty of some logical function of a node such as the physical radio interface, Media Access Control (MAC) Layer or Application Layer.

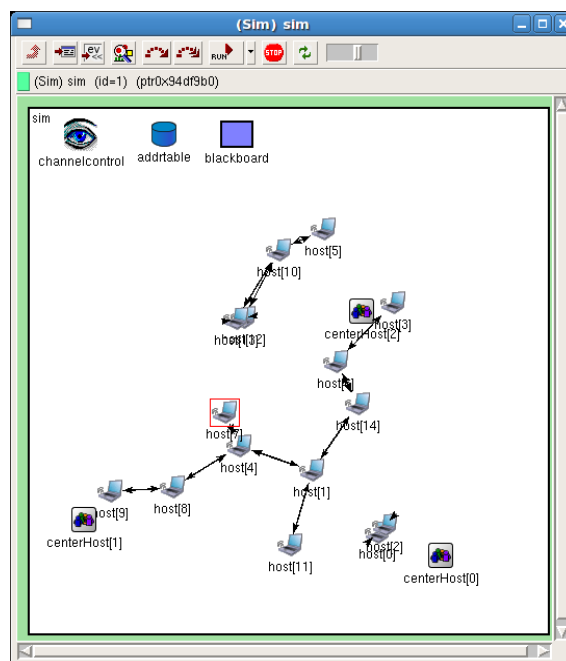


Figure 28: Sample simulation setup

Figure 28 shows the general setup of a scenario in the simulator. The system consists of several components described below:

- The *ChannelControl* module is responsible for handling all possible connections between the nodes. It does this in co-operation with the *mobility* module of each host module which in turn handles the movements of the node and communicates them to the channel control.
- The *addrtable* module does not model any real part of the system. It is used for collecting statistics from simulation runs and provides extra debugging information within the simulator.
- The global *blackboard* module provides an efficient way of exchanging information between layers without a need for passing pointers to different involved modules. The information which could be useful to other modules can be published on the blackboard and then it is

available to any subscribed parties. This feature has not only been used in several modules of the Mobility Framework but also in the implementation of the interface between the service framework and the *situation* module realized by VTT, which will be discussed later. Also the *Reference Point Group Mobility Model with Dynamic Groups* (described in section 6.15.3) utilizes the blackboard.

The *centerHost* modules are of a pseudo type; they only act as reference center points for the groups of nodes to follow and do not participate in network traffic in any way.

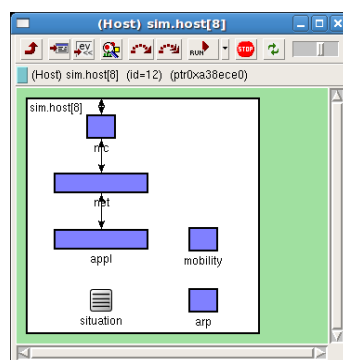


Figure 29: Modular structure of a node

The internal structure of a node in the simulator can be seen in figure 29. The different functional modules are represented as blue rectangles and the gates transferring packets between the layers are drawn as arrows. Individual modules which are not strictly part of the network layer stack are not connected by gates.

Starting from the top of the protocol stack, the module which simulates the service framework running on the node is called '*appl*'. The name follows the naming conventions of Mobility Framework and in our case it does not imply the kind of functionality it implements but instead it models the service layer.

The next module named '*net*' handles the actual network traffic by receiving and transferring data from/to the service layer according to the current situation. The radio interface management and simulation is done in the '*nic*' module. Each host handles its movement individually and it is done in the *mobility* module.

The *situation* module keeps track of the current network situation of the node and provides an interface to the service framework enabling it to subscribe. The events published on the interface gives the service layer the ability to obtain an abstract picture of the communication situation it is in at the moment and enables it to adapt its behavior accordingly. The situation module will also have an influence on how traffic is routed given the current situation and the quality of service demands received from the service framework. Finally, the *arp* module provides the MAC protocol as necessary.

Parameter	Value
Area size	500 m x 500 m
Nodes speed	10 m/s
CenterNodes speed	8 m/s
Bitrate	2 Mb/s
Carrier frequency	2.4 GHz
Max transmission power	4 mW
Signal attenuation threshold	-110 dBm
Path loss coefficient alpha	4
MAC protocol	ALOHA
Bits per control message	150
Bits per data message	1000
Number of runs per scenario	100

Table 2: Simulation parameters

7.3 Simulation scenarios

When viewed from the Wellness scenario [79] point of view, the mobility of users can be modelled, for example in two simulation scenarios. The first one models the use case when the network traffic flows inside relatively small groups of nodes. The above-mentioned refers to the situation in Wellness, where the service needs to communicate with the user's friends located nearby.

The second simulated use case could happen when the competitors of the cycling race gather to the service area of the competition prior to the race and the user has a need to communicate securely with a certain old friend. For this case, the Random Way Point mobility patterns suits most appropriately among the available models.

For the simulations, the default values found from table 2 were used if not otherwise mentioned.

7.4 Simulations carried out and preliminary results obtained

As the basis of the code required for the situated adaptive routing framework has been implemented, a few simple simulations have been executed to compare the amount of network traffic generated when the nodes use the situated adaptive opportunistic routing versus a plain epidemic routing protocol described in [80]. As the amount of devices acting in the BIONETS network can increase to considerably high numbers, the importance of avoiding unnecessary transmissions is of essence. For this reason, we chose to measure network usage and how it could be scaled down by allowing the nodes to be aware of their situation. The simulation set represented below focuses on one subset of basic network situations but future work will include more comprehensive situation handling and different usage scenarios.

The use case of the simulation explored below is when a U-node needs to communicate with another U-node. Depending on the service, it could also be that the message is restricted to be opened by only one node. In today's world many services need to be secured and therefore communication between parties is encrypted, so transferring the message to its recipient while minimizing the generated network usage seem to be essential. The mobile groups formed by the

moving users for example in vehicles and public events provide a good model and pattern where future BIONETS service could be used. However, some of the traffic will happen between two parties of U-nodes (and possibly T-nodes via their local U-node) and as our simulations show, it can be handled with smaller amounts of messages than with plain epidemic routing.

The epidemic routing implemented works by using a three-way handshake which the authors of the protocol call an anti-entropy session. The session is started between two nodes (A and B) which have not contacted each other within a defined time period. Once started, the node with a smaller identifier (now A) sends a summary vector to the other host (now B) where it lists in a compact way all the messages it has heard and wishes to advertise them to other nodes. Node B will then make a comparison of its own summary vector and the received one. The set difference of A's and B's vectors are sent back to A as a response indicating the desired messages. Upon receiving the response, the actual data messages can be transferred from A to B.

The situated adaptive routing has a module which will provide knowledge of different communicational roles dynamically assigned to nodes, restrictions and probable resources set by the hardware of the node and characteristics of the network group surrounding the node at the moment. Currently, the implementation in the simulator supports the possibility for different role awareness held by the nodes in the vicinity of the U-node. Also their identifiers are known to the situation module. Now, the identifiers should not be considered to be, e.g. location-dependent addresses. Instead, identifiers are location-independent and their purpose is to uniquely separate the different U-nodes from each other.

Let us consider a case where a source node sends information to some other defined node. The service running on the node might have got the identifier of the destination perhaps from some earlier engagement with the user of the node or by using a service request on the group of nodes the source is located at at the moment. Now the service can send the message down the protocol stack to the network layer. There the message's context-data, which has been set by the service, indicates that this is a normal priority message. Before the decision has been made how to forward and disseminate the message, the situation module can be queried about the roles and identifiers of near-by nodes. At the moment, the roles are to be used in the simulator more thoroughly. If a suitable identifier is found to be near, the message can be forwarded without making other copies, thus saving valuable resources both of the sender and of other nodes.

If the answer from the situation module is that no direct neighbor is the best candidate to receive the message, the situation module will send a query to its neighbors to determine the capabilities available in their proximity. Again, if a suitable node has been found with either the right identifier or the role to pass the message to, the situation module will instruct to send the message that way. In cases where no better alternative has been found, the fall back method in the simulator is that the message will be added to the heap of messages to be disseminated with the plain epidemic routing. Different kinds of queries and fall backs will be implemented on the simulator accordingly to the messages context-data and the resources of the node.

The simple optimizations on the network usage of information dissemination can lead to good improvements in the overall behavior of the system. Figures 30 and 31 show how the total amount of bytes generated in the network depends on the information dissemination used and the num-

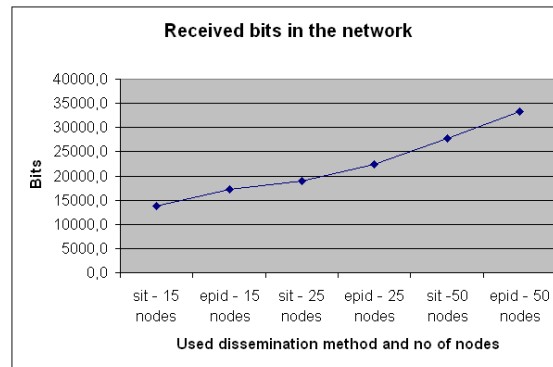


Figure 30: Total amount of received bits in the network vs. dissemination method vs. number of nodes

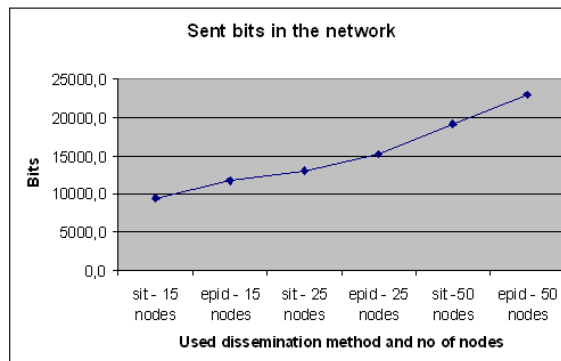


Figure 31: Total amount of sent bits in the network vs. dissemination method vs. number of nodes

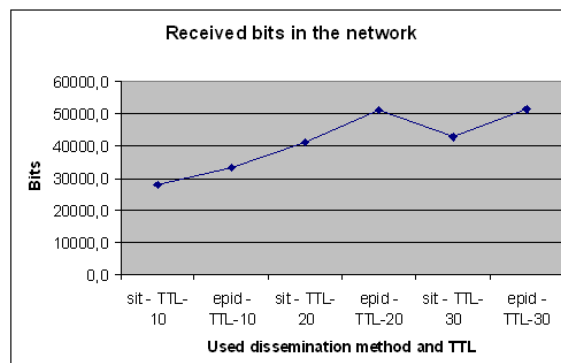


Figure 32: Total amount of received bits in the network vs. dissemination method vs. TTL

ber of nodes present. The mobility model utilized was GroupMobility and in every case the number of groups was five, so the size of a group was altered by changing the total amount of nodes. Other parameters of the network can be found in section 7.3. As it can be seen, the slopes at which the received and sent bits rise are the same in both figures. When using situated adaptive opportunistic routing, it saves transmissions regardless of the amount of nodes involved in the network. If 15, 25 and 50 nodes are considered, the amount of savings is of the same proportion

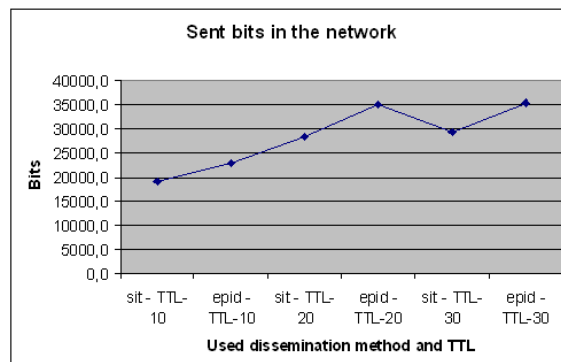


Figure 33: Total amount of sent bits in the network vs. dissemination method vs. TTL

when the nodes are increased in the network.

Figures 32 and 33 describe the relation between different dissemination methods and used time-to-live of the message against the total network traffic. In these cases, the amount of nodes was kept at static 50. The usage of TTL comes from the restriction that in BIONETS the clusters of nodes can be big, very mobile and the connections are of local type, i.e. the network is not connected as in mobile ad hoc networks (MANETS). The nodes possibly do not have in their possession a group identifier which would be known by every node forming the cluster. Therefore the flow of traffic must be filtered to avoid congestion and here it is dealt with by using TTL-fields in the packets. Figures 5 and 6 show that when the TTL is increased to 30, the amount of traffic raises only a small fraction when compared to TTL 20. Still it can be seen that the optimization works and keeps saving transmissions when compared to plain epidemic routing.

7.5 Conclusions and future work

As described earlier, even with simple knowledge of the situation the node is in, network routing can be optimized. While some portion of the network traffic in the future BIONETS network will be generated to share the environmental data coming from the T-nodes to a large number of U-nodes located near-by both in time and space, some of the traffic will be due to different services running on U-nodes and possibly behind access points. They might have use for a class of messages which are valuable only for the set destination and sender. These results indicate that kind of traffic could be optimized from purely epidemic types of dissemination. If more elaborate decisions regarding these situations are made when sending messages in the BIONETS network and more of these simple ones are also used together by introducing roles for nodes and categories for messages, the total amount of network traffic could be decreased.

The aim of VTT is to enhance the situated opportunistic routing algorithm in the simulator to enable more complicated routing situations and message classes, handle different kinds of arising application scenario requirements which arise, and interoperation with the legacy Internet networks in the future (see also D1.2.2). The aim is also to make a more detailed performance and scalability analysis when the number of nodes and heterogeneity will be increased. In addition, it looks valuable to evaluate by means of simulations the possible interoperation of situated adap-

tive opportunistic communication with some relevant related parts of a service framework, and enhance the realized simulation modules of situated opportunistic routing into smaller and more re-usable modules.

8 Message spreading in non-cooperative environments

In the dynamically deployed BIONETS wireless network,

where there is no centralized control of the nodes that participate in it,

the degree of cooperation among nodes may detrimentally affect the performance of the employed dissemination mechanisms.

Thus, it is important to evaluate the effect of the degree of cooperation on the efficiency of dissemination

mechanisms that have been designed under the assumption of fully cooperative nodes and to design

appropriate enhancements that would make these dissemination mechanisms capable of sustaining their efficiency under different degrees of cooperation.

In this section, we explore a simple mode of how information dissemination algorithms

could be adjusted to a non-cooperative environment by

assuming that each node has knowledge about the degree of

cooperation of the other nodes. We investigate the

effectiveness of a simple strategy that tries to avoid the

nodes that are less cooperative than a pre-specified threshold.

This complements the study reported in the previous deliverable

(D1.3.1, [81]), where the effects of cooperation on

the performance of three representative

algorithms were shown. In particular, the performance of the

epidemic (where every node forwards its message copies to the

node it encounters), the two-hop relay (where only the source

is allowed to spread up to a maximum number of message copies

to others, while the other nodes may keep their copy until

their encounter with the destination node) and the binary

spray-and-wait information dissemination algorithm (where every node

gives half of its message copies to every node with no copy it

encounters until it has only one copy – to give it to the

destination node) was studied within the context of a

non-cooperative environment.

The epidemic algorithm allows the fastest data delivery in

all cases (independently of whether the nodes cooperate or

not) as expected; however, the costs of employing the epidemic

algorithm are exorbitant. Binary spray-and wait performs

well (in comparison with the two-hop relay) in high degree of

cooperation environments since it achieves high dissemination

speed because of the multiple nodes that manage multiple copies. At the same time, as the cooperation degree of the nodes participating in the spreading process decreases, the probability of losing a large portion of the message copies that are totally allowed to be spread throughout the network increases. Furthermore, the message delivery delay reaches a higher level than that achieved by the two-hop relay algorithm (at the same time, however, transmissions are saved and less overhead is consumed). In conclusion, the two-hop relay algorithm seems to be the least sensitive one in terms of node cooperation, since the source itself is responsible for relaying the messages and no intermediate node may be in charge of more than one message copy in the network.

8.1 Adjustability to non-cooperative environments

It is assumed that there are two types of cooperation. A node may either drop a message upon reception (expressed by the dropping probability P_{drop} and capturing the effects of node misbehavior or the node's inability to store the message due to buffer limitations, Type I cooperation), or keep the message but avoid some forwarding transmissions that it is supposed to make according to the algorithm (expressed by the forwarding probability $P_{forward}$. This captures the effects of a less aggressive node misbehavior in comparison with Type I or the node's inability to afford the energy for all the necessary transmissions due to energy constraints, Type II cooperation).

More specifically, we assume that each node i drops the copy with probability $P_{drop,i}$ (Type I cooperation) or forwards it on behalf of some other node with probability $P_{forward,i}$ (Type II cooperation) and that this probability is known to the other nodes. We do not make any specific assumption about how the nodes acquire this information; this knowledge may become available through some reputation scheme (where the nodes' past actions may be monitored, recorded and then used to determine whether or not to forward the message) that works ideally and, thus, the exact values of $P_{drop,i}$,

and/or $P_{forward,i}$, are determined. Alternatively, we may assume that the nodes inform their neighbors about their degree of cooperation ($1 - P_{drop,i}$ for Type I cooperation, $P_{forward,i}$ for Type II cooperation); this would make sense, for example, in the case where the degree of cooperation of each node is proportional to its resources in a network of heterogeneous nodes.

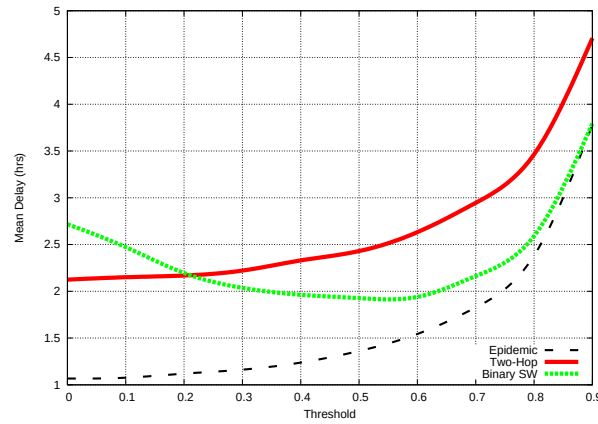
Provided that a node is aware of the degree of cooperation of each node it encounters, an upcoming issue is whether the node should avoid transmitting a copy of the message to nodes that will drop it with a high probability, for Type I cooperation (or forward the message with a small probability, for Type II cooperation). By avoiding the most non-cooperative nodes, one would expect to save some transmissions at the cost of a small increase in the delivery delay.

We investigate this idea by setting a threshold based on which every node decides on forwarding a message copy or not; that is, the node will give a copy of its message to node i only if $(1 - P_{drop,i})$ exceeds a threshold, for Type I cooperation (or if $P_{forward,i}$ exceeds a threshold, for Type II cooperation). For delay-sensitive traffic, this threshold could be a function of the time elapsed since message generation, since the more the elapsed time, the more it is worth taking the risk of giving the message to some node with a high P_{drop} (or small $P_{forward}$).

In Figure 8.1, the mean delivery delay of the three algorithms is illustrated as a function of the applied forwarding threshold in the case of Type I cooperation. The total number of transmissions induced is illustrated, for the three algorithms, in Figure 8.1. The corresponding results concerning Type II cooperation are depicted in figures 8.1 and 8.1, respectively.

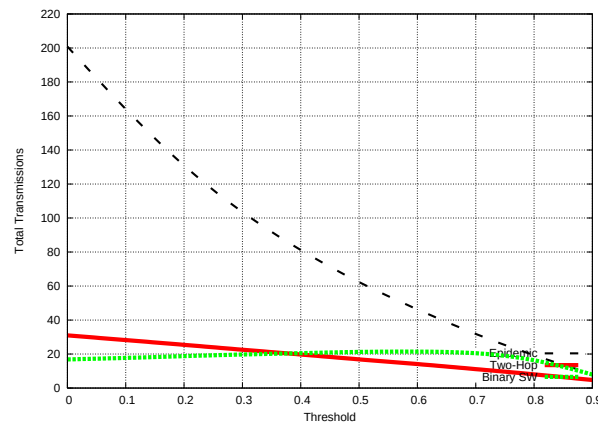
The results above refer to the case of a network of 100 nodes that have a degree of cooperation ranging from 0 to 1 (except for the source and destination nodes, which are assumed by default as cooperative), such that the mean degree of cooperation throughout the network is 0.5.

When no threshold is applied, the two-hop relaying algorithm achieves a lower delivery delay than the binary spray and wait in



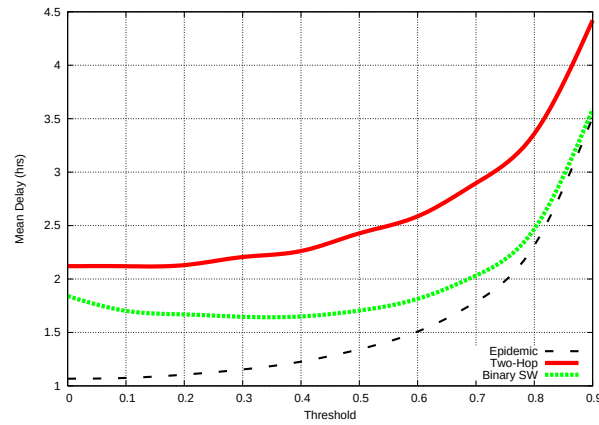
a function of the applied forwarding threshold (Type I cooperation)

Figure 34: The mean delivery delay of the three algorithms as a function of the applied forwarding threshold (Type I cooperation)



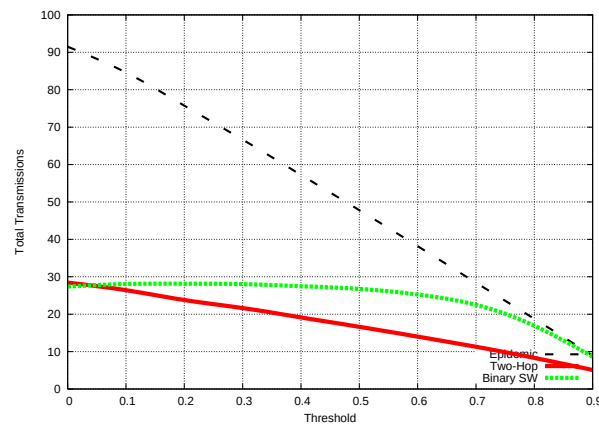
algorithms as a function of the applied forwarding threshold (Type I cooperation)

Figure 35: The total number of transmissions of the three algorithms as a function of the applied forwarding threshold (Type I cooperation)



a function of the applied forwarding threshold (Type II cooperation)

Figure 36: The mean delivery delay of the three algorithms as a function of the applied forwarding threshold (Type II cooperation)



algorithms as a function of the applied forwarding threshold (Type II cooperation)

Figure 37: The total number of transmissions of the three algorithms as a function of the applied forwarding threshold (Type II cooperation)

case of Type I cooperation. For Type II cooperation, the behavior of each algorithm seems to resemble the one in fully-cooperative environments. For both types of cooperation, the employment of the forwarding threshold leads to a significant reduction in the number of transmissions at the cost of only a small increase in the delivery delay, at least for a threshold less than the average degree of node cooperation. The only exception holds for the binary spray and wait algorithm in the case of Type I cooperation for which the inverse behavior is observed. More specifically, there seems to be a value of the threshold selected (~ 0.2 for the mean delay, and ~ 0.3 for total transmissions) below which the two-hop relaying algorithm achieves a lower delivery delay and/or induces more transmissions than the binary spray and wait, and vice versa, above that threshold. This may be justified by the fact that for relatively smaller values of the forwarding threshold, relatively more non-cooperative nodes will get and eventually drop a copy of the message. A node in binary spray and wait can be responsible for spreading up to $N/2$ message copies, thus the mean delay is seen to be worse than that achieved under the two-hop relaying algorithm, where the same non-cooperative nodes lose at most one copy. For this reason, and for the same values of the threshold, more transmissions take place under the two-hop relaying algorithm. For relatively higher values of the threshold, on the other hand, relatively more cooperative nodes are included in the spreading process; thus, the advantage of the spreading speed of the spray and wait algorithm (that clearly exists in a cooperative environment) starts to become evident.

Regarding Type II cooperation, the behavior of the algorithms resembles more the one in a fully cooperative environment. More specifically, the binary spray and wait algorithm always achieves a smaller delivery delay than the two-hop relaying algorithm but at the cost of a greater number of total transmissions. This may be justified by the fact that since message copies are not dropped, the binary spray and wait spreads the copies faster and, consequently, more copies are expected to have been spread by the time all copy carriers are notified of the message delivery.

9 Tnode to Unode communications: from real-world observation to a realistic model for the BIONETS simulation framework

As is highlighted in [82] unreliability of transmissions during contact times arises as a compelling issue in BIONETS environments. Contact times occur when two nodes enter on another's communication range and can communicate directly over a single-hop link. It is common use, when dealing with data-forwarding issues, to assume that contact times last enough to allow the data forwarding protocol to exchange all the data needed. However, depending on the particular application scenario, both the amount of data which needs exchanging and the number of nodes willing to communicate during the same contact time can be quite variable. Hence, it is necessary to investigate how transmissions during contact times are carried out. Specifically, this kind of investigation needs focus on both the *feasibility* of reliable data exchange during contact times, and on the strategies to maximize the *throughput* experience during data exchange.

9.1 Motivation and Problem Statement

Communications during contact times must be extremely *fast* and *effective*. This requires that *novel communication paradigms* be conceived which take into account the absence of stable links and exploit the occurrence of even short contacts. Chatty communication paradigms are not suitable for environments where contact durations are not predictable [83], like BIONETS environments. Among these protocols there are, for example, acknowledge-based protocols, in which the sender node waits for acknowledgements to know if the data it has sent has actually arrived at destination, and to manage possible retransmissions. Better performance can be envisaged when using communication protocols which minimize synchronization between the communication peers and that provide all the data together ('one-way') in a single transaction. In particular, communication paradigms based on encoding techniques [84] seem quite suitable. They have the potential to consume a shorter portion of contact time because the source node needs to simply transmit the generated codes without waiting for any acknowledgement. The contact time is therefore used only for transmission of information content rather than for control data. However, depending on the amount of redundancy which is added to the original data, part of the contact time can be wasted with encoding-based communication paradigms as well. This happens, for example, when too much redundancy is generated and part of the contact time is used to transmit useless codes which the destination does not need because it is already able to decode. This can cause bandwidth wastage and increase the delay suffered between transmissions of consecutive bursts of data.

Encoding-based communication paradigms can be applied to both Unode to Unode communications and Tnode to Unode communications. In the first case they can improve network performance both in terms of *delivery success rate* and *throughput* achieved. The delivery success rate increases since all contacts (despite their shortness) can successfully be exploited to exchange data. The throughput instead increases because larger amounts of data can be exchanged per

time unit. In the case of communications from Tnodes to Unodes, encoding-based communication paradigms serve for context-data retrieval. This may be more or less challenging depending on the amount of context data that Tnodes store. The BIONETS architecture includes different types of Tnodes, some of which have the potential to deal with a large amount of data. However, encoding-based communication protocols are not only suitable for Tnodes which deal with large amounts of data. The fact that minimum synchronization is required between Tnodes and Unodes for data acquisition is especially beneficial in case multiple Unodes visit the same Tnode simultaneously. They all can benefit from the same transmission from the Tnode without wasting part of the contact time waiting for the Tnode to have finished the transmission to the previous Unode. Another interesting case arises when multiple Tnodes are densely deployed in the area under study and collect the same data. Hence, when a Unode passes it can take part of the context data from a Tnode and part from another Tnode that becomes visible when the previous one is no longer visible. Passing by the second Tnode the Unode is finally able to reconstruct the entire context data.

At this stage of the simulation workpackage we focus on the particular case of communications from Tnode to Unode and introduce a model for Tnodes to Unodes communications in the BIONETS simulation framework. This model will allow for simulation of the data acquisition process at a Unode when it encounters a Tnode enroute. This will lead to many realistic simulations and will also introduce cases in which the acquisition of context-data fails due to the occurrence of particularly disadvantageous conditions like contact times which are too short. To build a realistic data acquisition model we conducted a real-world observation through an experimentation campaign, which is described in [82]. Results from the experimentation showed, as was expected, that an encoding-based approach allows better exploitation of limited contact-time resources. However, to allow for better performance it is necessary to adapt the redundancy level to the actual communication environment, and particularly to the degree of packet loss which is experienced. Based on these observations we have developed a novel adaptive communication protocol based on encoding, which we briefly describe in Section 9.1.2.

Results from this investigation are also promising in terms of Unode to Unode communications because Unodes are less resource-constrained, thus it would be easier to apply such techniques to them. However, new issues arise when using Unodes, particularly concerning interaction with the data dissemination protocol adopted at the Unode plane. Hence further and special purpose investigation is needed.

9.1.1 Application Scenarios

In this section we consider one of the BIONETS scenarios which are proposed and analyzed in [85] and we point out how encoding-based communication protocols can make gathering of context information more robust and reliable in this scenario.

In the *Virtual Guides* application scenario, a virtual guide is embedded in an urban pervasive environment to help citizens in their every-day lives and tourists in their cultural visits. Among

the pieces of information that users can gather from the Tnodes deployed in the city there are the city's places of interest (e.g., historical buildings, cinemas, and restaurants), and directions, or even complete city maps, to reach particular places. Furthermore, context information can assume different forms depending on both its nature and the device used as a Unode by users. It can be simple textual data (e.g. advertisements, architectural history, and state of reservation transactions), images (e.g. street maps), or even videos (e.g. visual guides and movie trailers). In these last cases, a massive quantity of bytes needs to be transmitted from the Tnodes plane to the Unodes plane, and it can take special advantage from adoption of encoding-based communication protocols. Benefits are also evident depending on the particular data acquisition pattern that occurs. Unodes may be carried by tourists walking from place to place and gathering data from the Tnodes encountered near monuments or museums, but they can also be carried by cars with people looking for directions or city maps. This case is a bit more challenging because transmissions to a running car can exploit only very small contact times, while the amount of data exchanged can be reasonably high. Furthermore, what happens when multiple cars pass almost simultaneously near the same Tnode and they all try to gather information from it? The Tnode should alternate transmissions of the data it stores among the different cars. Moreover, given that transmissions to a mobile car suffer from high packet loss, the Tnode should also deal with different independent retransmissions towards the different requesting cars. This would consume a long portion of contact time and lead to the risk that none of the cars receives all the data needed. In case codes are transmitted instead of plain data, all the cars can catch useful data from the same transmission like in a multicast session. Whenever a code gets lost, a car just needs to catch another one. No independent retransmissions need to be managed, cars are served at the same time and the contact time is exploited most efficiently. In a possible alternative data gathering pattern a car can exploit information from different Tnodes encountered in sequence. This may happen when the contact time between the moving auto and a single Tnode met is too short. Hence, the car starts receiving the first set of codes from a first Tnode encountered, but when it passes beyond the Tnode it is not able to decode the information because it has not caught enough codes. When soon after the car encounters a subsequent Tnode it can catch other codes with which it is finally able to do the decoding. Similar data acquisition patterns can also be thought when Unodes move slower than cars. We can envision a group of tourists visiting a museum and receiving data from Tnodes located beside paintings or sculptures. The data can be the voice of a tourist guide giving historical information about the painting or sculpture, thus we would have a considerable amount of data transmitted and multiple destinations waiting for it. In this case the contact time is not that critical because tourists typically stay in one place for a while but the management of multiple transmissions towards multiple receivers would be the more critical issue to tackle.

Similar considerations on i) the amount of data exchanged during single contact times, on ii) the relative speed of the communicating nodes which causes a high degree of packet loss, and on iii) the data acquisition patterns which can be disadvantageous to reliable transmissions, are also valid for other BIONETS application scenarios, like the *Wellness/Fitness* scenario. However, such discussion here would not add further information to this dissertation.

9.1.2 A novel adaptive communication paradigm based on Reed-Solomon codes

We developed a novel communication protocol that adapts encoding to the actual communication environment. Our protocol is based on Reed-Solomon (RS) codes and tunes encoding over the packet loss registered during transmissions. In fact, it generates new codes when the packet loss experienced is high while only a minimum set of codes is produced when the packet loss is low. Further details on our communication protocol can be found in [82]. We tested our protocol in a real testbed to validate its efficacy and *sampled* the arrival times of packets (codes) to the mobile node. The resulting traces describe the arrival process of data at the Unodes while passing in proximity of Tnodes. The integration of these traces into the BIONETS simulation framework allows the framework to have a realistic view of the ability of Unodes to collect context data from the surrounding environment. Whenever a Unode approaches a Tnode, a different data acquisition pattern is triggered depending on the speed of the Unode and on its distance from the Tnode. This allows the BIONETS simulation framework to distinguish between cases in which data acquisition is successful from cases where it is not successful due to a large amount of data or a too small contact time.

In the next sections we discuss the integration of our traces into the BIONETS simulation framework.

9.2 Tnode to Unode Communications Modelling

9.2.1 Tnode to Unode communications in the BIONETS simulation framework

The goal of the BIONETS simulator is to develop a useful instrument to be used during the entire project lifetime to test and validate the ideas proposed in the project. To this end, the most challenging task was to design a flexible modular architecture that allowed for the easy adding of other modules modelling new ideas emerging later on in the project.

The current stage of the BIONETS Simulator [86] consists only of the architecture of one of the most complex entities of the BIONETS system: the Unode. Tnodes are not currently being designed. As a consequence, only communications among Unodes are taken into account. This also implies that, presently, the BIONETS simulator allows only for the validation of solutions for this kind of communications like information dissemination algorithms or active networking.

In the next phase of the evolution process of the simulator, the data gathering process for the context information needed by the Unodes to run services will be integrated in the simulator. This goal can be achieved without necessarily requiring the complete design of the Tnode architecture. Instead, communications among the Tnode plane and the Unode plane can successfully be integrated into the BIONETS simulation framework by means of a traffic generator which produces data according to a realistic pattern which is described by real-world traces.

The traffic generator provides context data to the Unode with the same frequency described by the traces. As previously explained, the integration of such real-world traces provides the simulator with a more realistic view of the Tnode to Unode communications.

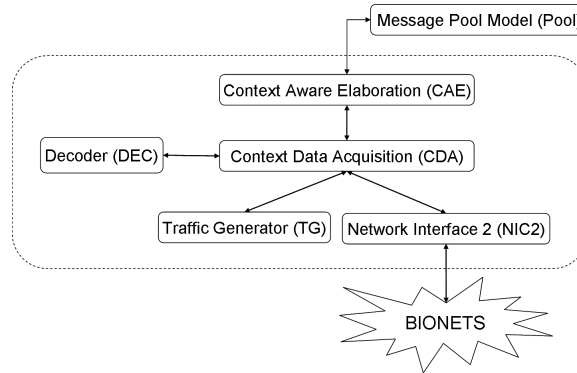


Figure 38: Part of the Unode architecture modelling the Tnodes to Unodes Communications

9.2.2 Trace-driven simulations

In order to allow trace-driven simulations into the BIONETS simulator, it is necessary to efficiently integrate the module that describes real traces, namely the **Traffic Generator (TG)**, into the architecture itself [86]. To this aim, new additional modules are required to better manage the communication between the TG module and the other existing modules that compose the original Unode architecture [87]. This guarantees that each Unode is not only able to receive encoded data passed by the generator of real traces but it can also interpret and use it appropriately.

The overview of the part of Unode architecture that models the Tnode to Unode communications is shown in Figure 38. It consists of five main modules with different tasks. In the following, a brief description of each module is given.

NETWORK INTERFACE 2 (NIC2): This is the network interface of the Unode that models communication among Unodes and Tnodes. Since Tnodes have been grouped to form four classes (see [84]), the NIC2 should model several communications technologies (e.g., RFID, 802.15.4, 802.11).

TRAFFIC GENERATOR (TG): The Traffic Generator module is responsible for providing traces describing the real representation of the inter-arrival times of the data at the Unode, as described in the previous paragraph.

CONTEXT DATA ACQUISITION (CDA): The Context Data Acquisition module triggers the communication with adjacent Tnodes and queries them to gather context information. This can be done using one of the two models proposed in [84]: Push or Pull. Depending on the chosen model, CDA is able to listen to signals coming from the environment (i.e. advertisements sent by Tnodes), or send explicit requests for a particular service. In addition, when the context information is encoded, it also regulates the coordination between TG and DEC.

CONTEXT AWARE ELABORATION (CAE): The task of the Context Aware Elaboration is to elaborate the context information gathered from the environment before passing it to the service or storing it in the Message Pool. This also implies that it sends requests of context information when a service needs it.

DECODER (DEC): The Decoder module is responsible for all the tasks connected with the

decoding phase. These include: *i)* recognize encoded blocks that are linearly independent, *ii)* discard blocks of encoded data that are not linearly independent, *iii)* decode the encoded data to reconstruct the original data, *iv)* ask for more blocks of encoded data when those which had arrived to the Unode are not sufficient to reconstruct the source data blocks.

Referring to the architecture in Figure 38, we have made some preliminary hypotheses about each module. More precisely, Unodes gather context information provided only by the Traffic Generator module. This implies that currently there is not real communication between Unodes and Tnodes through the NIC2. In addition, the CDA module implements only those features sufficient for coordinating TG and DEC. These choices guarantee the smooth running of trace-driven simulations in the early stages as well. Under such a hypothesis, when GT receives an explicit request of context information from the CDA, it starts providing blocks of encoded data according to specific inter-arrival times. They are passed to the DEC that starts the decoding procedure. When all the encoded blocks are successfully received, the original decoded data is finally passed to the CAE.

References

- [1] E. V. T. C. M. B. L. B. B. T. F. D. P. I. C. D. M. A. P. A. V. E. A. M. D. F. B. A. A. H. Vilmos Simon, Sándor Szabó, "D1.3.1 the initial mathematical models of new bionets network elements and algorithms," Tech. Rep., 2007.
- [2] M. M. B. Tariq, M. Ammar, and E. Zegura, "Message ferry route design for sparse ad hoc networks with mobile nodes," in *Proc. of ACM MobiHoc*, Florence, Italy, May 22–25, 2006, pp. 37–48.
- [3] K. Fall, "A delay-tolerant network architecture for challenged internets," in *Proc. of ACM SIGCOMM*, Karlsruhe, Germany, March 25–29, 2003.
- [4] J. Leguay, A. Lindgren, J. Scott, T. Friedman, and J. Crowcroft, "Opportunistic content distribution in a urban setting," in *Proc. of ACM Chants*, Florence, IT, September 15, 2006.
- [5] J. Burgess, B. Gallagher, D. Jensen, and B. N. Levine, "Maxprop: Routing for vehicle-based disruption-tolerant networking," in *Proc. of IEEE INFOCOM*, Barcelona, Spain, April 23–29, 2006.
- [6] U. Lee, E. Magistretti, B. Zhou, M. Gerla, P. Bellavista, and A. Corradi, "MobEyes: smart mobs for urban monitoring with vehicular sensor networks," UCLA CSD, Tech. Rep. 060015, 2006. [Online]. Available: <http://netlab.cs.ucla.edu/wiki/files/mobeyestr06.pdf>
- [7] M. Grossglauser and D. Tse, "Mobility increases the capacity of ad hoc wireless networks," *IEEE/ACM Trans. on Networking*, vol. 10, no. 4, pp. 477–486, Aug. 2002.
- [8] P. Gupta and P. R. Kumar, "The capacity of wireless networks," *IEEE Trans. on Information Theory*, vol. 46, no. 2, pp. 388–404, March 2000.
- [9] A. E. Gamal, J. Mammen, B. Prabhakar, and D. Shah, "Throughput-delay trade-off in wireless networks," in *Proc. of IEEE INFOCOM*, Hong Kong, China, March 7–11, 2004.
- [10] M. Neely and E. Modiano, "Capacity and delay tradeoffs for ad hoc mobile networks," *IEEE Trans. on Information Theory*, vol. 51, no. 6, pp. 1917–1937, 2006.
- [11] S. Jain, K. Fall, and R. Patra, "Routing in a delay tolerant network," *SIGCOMM Comp. Comm. Rev.*, vol. 34, no. 4, pp. 145–158, Oct. 2004.
- [12] W. Zhao, M. Ammar, and E. Zegura, "Controlling the mobility of multiple data transport ferries in a delay-tolerant network," in *Proc. of IEEE INFOCOM*, Miami USA, March 13–17, 2005.
- [13] A. Khelil, C. Becker, J. Tian, and K. Rothermel, "An epidemic model for information diffusion in MANETs," in *Proc. of ACM MSWiM*, Atlanta, Georgia, Sept. 28, 2002, pp. 54–60.
- [14] K. Harras, K. Almeroth, and E. Belding-Royer, "Delay tolerant mobile networks (DTMNs): Controlled flooding schemes in sparse mobile networks," in *Proc. of IFIP Networking*, Waterloo, Canada, May, 2005.

- [15] A. Chaintreau, P. Hui, J. Crowcroft, C. Diot, R. Gass, and J. Scott, "Impact of human mobility on the design of opportunistic forwarding algorithms," in *Proc. of INFOCOM*, Barcelona, Spain, April 23–29, 2006.
- [16] M. Demmer, E. Brewer, K. Fall, S. Jain, M. Ho, and R. Patra, "Implementing delay tolerant networking," Intel, Tech. Rep. IRB-TR-04-020, 28 Dec. 2004. [Online]. Available: <http://www.dtnrg.org/papers/demmer-irb-tr-04-020.pdf>
- [17] S. Merugu, M. Ammar, and E. Zegura, "Routing in space and time in networks with predictable mobility," Georgia Tech., College of Computing, Tech. Rep. GIT-CC-04-07, 2004. [Online]. Available: <http://citeseer.ist.psu.edu/merugu04routing.html>
- [18] L. R. Ford and D. R. Fulkerson, "Constructing maximal dynamic flows from static flows," *Operations Research*, vol. 6, no. 3, pp. 419–433, May-June 1958.
- [19] A. Ferreira, "Building a reference combinatorial model for MANETs," *IEEE Network*, vol. 8, no. 5, pp. 24–29, Sept./October 2005.
- [20] A. Ferreira and A. Jarry, "Complexity of minimum spanning tree in evolving graphs and the minimum-energy broadcast routing problem," in *Proc. of WiOpt'04*, Cambridge, UK, March 24–26, 2004.
- [21] M. Penrose, *Random Geometric Graphs*. N. Y.: Oxford University Press, 2003.
- [22] I. Carreras, D. Miorandi, and I. Chlamtac, "A framework for opportunistic forwarding in disconnected networks," in *Proc. of Mobiquitous*, Palo Alto, USA, July 17–21, 2006.
- [23] T. Camp, J. Boleng, and V. Davies, "A survey of mobility models for ad hoc network research," *Wiley Wireless Communications and Mobile Computing*, vol. 2, no. 5, pp. 483–502, August 2002.
- [24] N. Sarshar, P. O. Boykin, and V. P. Roychowdhury, "Scalable percolation search on complex networks." *Theor. Comput. Sci.*, vol. 355, no. 1, pp. 48–64, April 2006.
- [25] B. Bollobas, *Random Graphs*. Cambridge: Cambridge Univ. Press, 2001.
- [26] F. Chung and L. Lu, "The diameter of sparse random graphs," *Advances in Applied Mathematics*, vol. 26, no. 4, pp. 257–279, 2001.
- [27] R. Albert and A. L. Barabasi, "Statistical mechanics of complex networks," *Review of Modern Physics*, vol. 74, no. 1, pp. 47–95, Jan. 2002.
- [28] F. D. Pellegrini, D. Miorandi, I. Carreras, and I. Chlamtac, "A graph based model for sparse ad hoc networks," in *Proc. of IEEE INFOCOM*, Anchorage, USA, May 7–9, 2007.
- [29] J.-Y. L. Boudec and M. Vojnovic, "Perfect simulation and stationarity of a class of mobility models," in *Proc. of IEEE INFOCOM*, Miami, USA, March 13–17, 2005.
- [30] AA.VV., "Omnet++." [Online]. Available: www.omnetpp.org

- [31] The disruption tolerant networking project at UMass. [Online]. Available: <http://prisms.cs.umass.edu/diesel/>
- [32] A. Viterbi, *CDMA: Principles of Spread Spectrum Communication*, First ed. Addison-Wesley, 1995.
- [33] D. Tse and S. Hanly, "Linear Multiuser Receivers: Effective Interference, Effective Bandwidth and User Capacity," *IEEE Trans. on Information Theory*, vol. 45, no. 2, pp. 641–657, Mar. 1999.
- [34] S. Verdú and S. Shamai, "Spectral Efficiency of CDMA with Random Spreading," *IEEE Trans. on Information Theory*, pp. 622–640, Mar. 1999.
- [35] J. Evans and D. Tse, "Large System Performance of Linear Multiuser Receivers in Multipath Fading Channels," *IEEE Trans. on Information Theory*, pp. 2059–2078, Sept. 2000.
- [36] S. Shamai and S. Verdú, "The Effect of Frequency-Flat Fading on the Spectral Efficiency of CDMA," *IEEE Trans. on Information Theory*, vol. 47, no. 4, pp. 1302–1327, May 2001.
- [37] A. Tulino, L. Li, and S. Verdú, "Spectral Efficiency of Multicarrier CDMA," *IEEE Trans. on Information Theory*, vol. 51, no. 2, pp. 479–505, February 2005.
- [38] K. Gilhousen, I. Jacobs, R. Padovani, A. Viterbi, L. W. Jr., and C. Wheatley, "On the Capacity of a Cellular CDMA System," *IEEE Trans. on Veh. Technol.*, vol. 40, pp. 303–312, 1991.
- [39] B. Zaidel, S. Shamai, and S. Verdú, "Multicell Uplink Spectral Efficiency of Coded DS-CDMA with Random Signatures," *IEEE Journal on Selected Areas in Communications*, vol. 19, pp. 1556–1569, 2001.
- [40] M. Debbah, "Downlink CDMA: to Cell or not to Cell," in *12th European Signal Processing Conference (EUSIPCO 2004)*, Vienna, Austria, Sept. 2004.
- [41] N. Bonneau, M. Debbah, and E. Altman, "Spectral Efficiency of CDMA Downlink Cellular Networks with Matched Filter," *EURASIP Journal on Wireless Communications and Networking*, pp. 1–10, 2006.
- [42] N. Bonneau, M. Debbah, E. Altman, and G. Caire, "Spectral Efficiency of Uplink CDMA Cellular Networks," in *IEEE International Conference on Acoustics, Speech, and Signal Processing*, Philadelphia, USA, Mar. 2005.
- [43] A. Tulino and S. Verdú, "Random matrix theory and wireless communications," *Foundations and Trends in Communications and Information Theory*, vol. 1, no. 1, 2004.
- [44] P. Gupta and P. Kumar, "The capacity of wireless networks," *IEEE Trans. Inform. Theory*, vol. 46, no. 2, pp. 388–404, Mar. 2000.
- [45] A. Özgür, O. Lévêque, and D. Tse, "How does the information capacity of ad hoc networks scale?" in *Proc. 44th Annual Allerton Conference on Communication, Control, and Computing*, Sept. 2006.

- [46] H. Bolcskei, R. Nabar, O. Oyman, and A. Paulraj, "Capacity scaling laws in mimo relay networks," *IEEE Trans. Wireless Commun.*, vol. 5, no. 6, pp. 1433–1444, June 2006.
- [47] N. Ida, *Engineering Electromagnetics, 2nd Edition*. Springer, May 2004.
- [48] T. S. Rappaport, *Wireless Communications Principles and Practice, 2nd Edition*, ser. Communications Engineering and Emerging Technologies. Prentice Hall, 2002.
- [49] R. Gray, "Toeplitz and circulant matrices: A review," in *Foundations and Trends in Communications and Information Theory*. Stanford University: NOW, the essence of knowledge, 2006, vol. 2, pp. 155–239.
- [50] E. Altman, T. Boulogne, R. E. Azouzi, T. Jimenez, and L. Wynter, "A survey on networking games," *Computers and Operations Research*, 2006.
- [51] R. Yates, "A Framework for Uplink Power Control in Cellular Radio Systems," *IEEE Journal on Selected Areas in Communications*, vol. 13, no. 7, pp. 1341–1347, Sept. 1995.
- [52] D. Goodman and N. Mandayam, "Power Control for Wireless Data," *IEEE Personal Communications*, vol. 7, no. 2, pp. 48–54, 2000.
- [53] A. MacKenzie and S. Wicker, "Game Theory and the Design of Self-Configuring, Adaptive Wireless Networks," *IEEE Trans. on Communications*, Nov. 2001.
- [54] J. Wardrop, "Some Theoretical Aspects of Road Traffic Research Communication Networks," in *Proc. Inst. Civ. Eng.*, vol. 2, no. 1, 1952, pp. 325–378.
- [55] F. Meshkati, H. Poor, S. Schwartz, and N. Mandayam, "An Energy-Efficient Approach to Power Control and Receiver Design in Wireless Data Networks," *IEEE Trans. on Communications*, vol. 53, no. 11, pp. 1885–1894, Nov. 2005.
- [56] V. Rodriguez, "Robust Modeling and Analysis for Wireless Data Resource Management," in *Proceedings of the IEEE Wireless Communications & Networking Conference*, Mar. 2003.
- [57] F. Meshkati, M. Chiang, H. Poor, and S. Schwartz, "A Game-Theoretic Approach to Energy-Efficient Power Control in Multi-Carrier CDMA Systems," *IEEE Journal on Selected Areas in Communications*, 2006.
- [58] V. Girko, *Theory of Random Determinants*. Kluwer Academic Publishers, Dordrecht, The Netherlands, 1990.
- [59] J. Silverstein and Z. Bai, "On the Empirical Distribution of Eigenvalues of a Class of Large Dimensional Random Matrices," *J. Multivariate Anal.*, vol. 54, no. 2, pp. 175–192, 1995.
- [60] C. Saraydar, N. Mandayam, and D. Goodman, "Efficient Power Control via Pricing in Wireless Data Networks," *IEEE Trans. on Communications*, vol. 50, no. 2, pp. 291–303, Feb. 2002.

- [61] R. Müller and S. Verdú, "Design and Analysis of Low-Complexity Interference Mitigation on Vector Channels," *IEEE Journal on Selected Areas in Communications*, pp. 1429–1441, Aug. 2001.
- [62] A. Varga *et al.*, "The OMNeT++ discrete event simulation system," *Proceedings of the European Simulation Multiconference*, 2001.
- [63] —, "The OMNeT++ Discrete Event Simulation System," <http://www.omnetpp.org/>, 2007.
- [64] W. Drytkiewicz, S. Sroka, V. Handziski, A. Koepke, and H. Karl, "A mobility framework for omnet++," *3rd International OMNeT++ Workshop*, January, 2003.
- [65] H. K. *et al.*, "A mobility framework for omnet++," <http://mobility-fw.sourceforge.net/>, 2007.
- [66] B. Williams and T. Camp, "Comparison of broadcasting techniques for mobile ad hoc networks," 2002.
- [67] J. W. Fei Dai, "Performance analysis of broadcast protocols in ad hoc networks based on self-pruning," 2004.
- [68] Y.-S. C. Sze-Yao Ni, Yu-Chee Tseng and J.-P. Sheu, "The broadcast storm problem in a mobile ad hoc network."
- [69] F. D. Jie Wu, "Broadcasting in ad hoc networks based on self-pruning," 2003.
- [70] X. L. W. Peng, "On the reduction of broadcast redundancy in mobile ad hoc networks," *Proceedings of MOBIHOC*, 2000.
- [71] L. V. A. Qayyum and A. Laouiti., "Multipoint relaying: An efficient technique for flooding in mobile wireless networks," INRIA, Tech. Rep., 2000.
- [72] C. B.-K. R. Abdelmajid Khelil, Pedro José Marrón, "Hypergossiping: A generalized broadcast strategy for mobile ad hoc networks," 2006.
- [73] H. L. J. Wu, "On calculating connected dominating set for efficient routing in ad hoc wireless networks," *Proceedings of the Third International Workshop on Discrete Algorithms and Methods for Mobile Computing and Communications (DiaLM)*, 1999.
- [74] X. L. Ivan Stojmenovic, "Loop-free hybrid single-path/flooding routing algorithms with guaranteed delivery for wireless networks," 2001.
- [75] R. J. Vamsi K Paruchuria, Arjan Duresib, "Optimized flooding protocol for ad hoc networks."
- [76] S. K. Xiaohu Chen, Michalis Faloutsos, "Distance adaptive (dad) broadcasting for ad hoc networks."

- [77] X. L. W. Peng, "Ahbp: An efficient broadcast protocol for mobile ad hoc networks," *Journal of Science and Technology*, 2002.
- [78] V. D. Tracy Camp, Jeff Boleng, "A survey of mobility models for ad hoc network research."
- [79] T. B. project, "Service architecture: requirement specification and concept definition. Deliverable D3.1.1," pp. 9–10, 2006.
- [80] A. Vahdat and D. Becker, "Epidemic routing for partially connected ad hoc networks," *Duke University*, 2000.
- [81] V. Simon, S. Szabó, E. Varga, T. Csvórics, M. Bérces, L. Bacsárdi, B. Tujner, F. D. Pellegrini, I. Carreras, D. Miorandi, A. Panagakis, A. Vaio, E. Altman, M. Debbah, F. Baude, and A. A. Hanbali, "The initial mathematical models of new bionets network elements and algorithms," Bionets Deliverable (D1.3.1), December 2006. [Online]. Available: http://www.bionets.eu/docs/BIONETS_D1_3_1.pdf
- [82] Create-Net, "D1.2.2 Disappearing Network Autonomic Operation and Evolution," 2007, BIONETS, IST-2004-2.3.4 - FP6-027748. [Online]. Available: <http://www.bionets.org/>
- [83] V. Cerf, S. Burleigh, A. Hooke, L. Torgerson, R. Durst, K. Scott, K. Fall, and H. Weiss, "Delay-tolerant Network Architecture," 2005. [Online]. Available: <http://www.dtnrg.org/docs/specs/draft-irtf-dtnrg-arch-03.txt/>
- [84] F. D. Pellegrini, D. Miorandi, I. Carreras, G. Alfano, D. Raz, R. Cohen, E. Borgia, J. Latvakoski, D. Schreckling, A. Panagakis, and A. Vaio, "Disappearing network infrastructure and design: Functionality and challenges," Bionets Deliverable (D1.2.1), September 2006. [Online]. Available: http://www.bionets.eu/docs/BIONETS_D1_2_1.pdf
- [85] VTT, "D3.1.1 Service Architecture: requirement specification and concept definition," 2006, BIONETS, IST-2004-2.3.4 - FP6-027748. [Online]. Available: <http://www.bionets.org/>
- [86] "Bionets simulator project official website (bionets.hit.bme.hu)." [Online]. Available: bionets.hit.bme.hu
- [87] "BIONETS SIMULATOR MODULE Library Architecture design v0.1," 2007, BUTE, <http://bionets.hit.bme.hu>.